Trigonometry

Questions 1–325

2.01 — Question 1. Copy and complete the following table:

Degrees	Radians
180°	
90°	
	$\frac{\pi}{3}$
	$\frac{\pi}{4}$
	$\frac{\pi}{5}$
30°	
20°	
	$\frac{8\pi}{9}$
1.62°	
	1.62

2.01 — Question 2. Find the length of each of these sector's arc with angle θ radians and radius r:

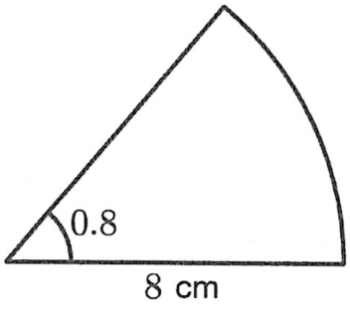
a) $\theta = 0.5$, $r = 10$ cm
b) $\theta = \frac{\pi}{4}$, $r = 20$ m
c) $\theta = 0.75$, $r = 15$ cm
d) $\theta = \frac{3\pi}{8}$, $r = 120$ m

2.01 — Question 3. Find the area of each of these sectors with angle θ radians and radius r:

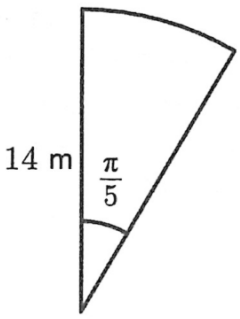
a) $\theta = 0.4$, $r = 20$ cm
b) $\theta = \frac{\pi}{3}$, $r = 9$ m
c) $\theta = 1.15$, $r = 64$ cm
d) $\theta = \frac{\pi}{20}$, $r = 18.5$ m

2.01 — Question 4. Find the area and perimeter of the following sectors, given that the angle shown is measured in radians:

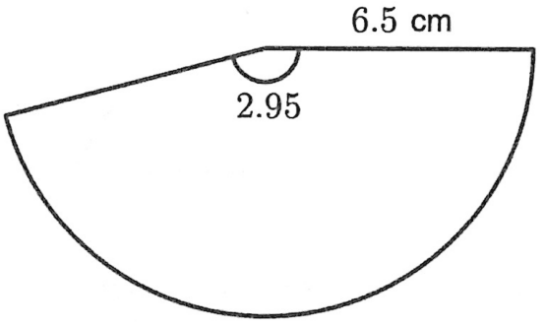
a)



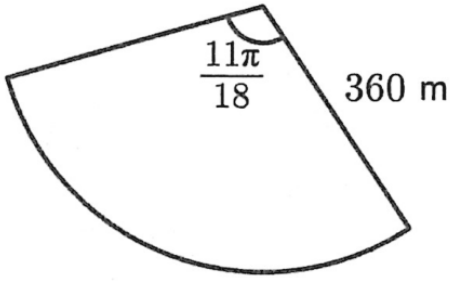
b)



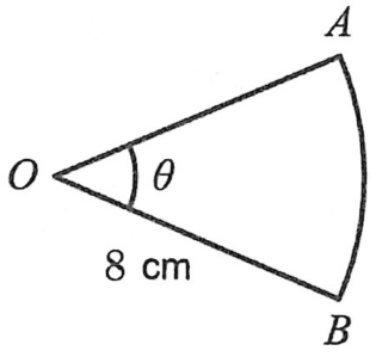
c)



d)



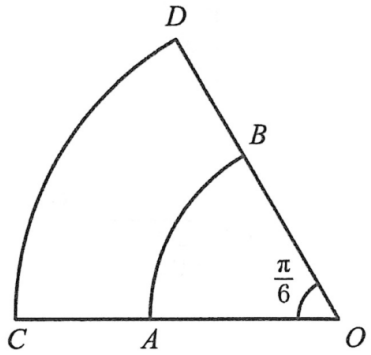
2.01 — Question 5. A sector of a circle, OAB, with radius 8 cm has perimeter 25 cm. The angle AOB is θ radians. Find the value of θ.



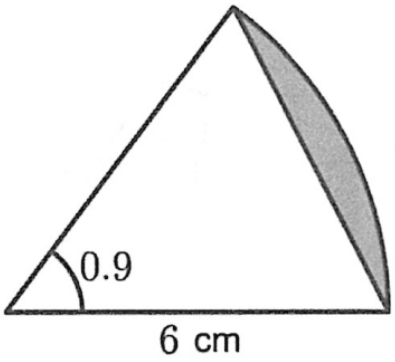
2.01 — Question 6. A sector of a circle with radius r cm and angle θ radians has an area of 50 cm^2. Write down an equation for r in terms of θ

2.01 — Question 7. A sector of a circle with radius r cm and angle θ radians has a perimeter of 84 m. Write down an equation for θ in terms of r

2.01 — Question 8. The diagram below shows two sectors, OAB and OCD. OAC and OBD are straight lines. The angle AOB is $\frac{\pi}{6}$ radians. The length of OC is 9 cm. It is given that the area of sector OCD is three times larger than the area of OAB. Find the exact length of OA.

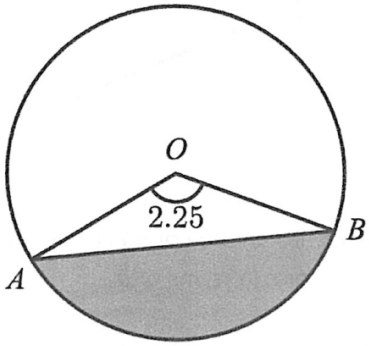


2.01 — Question 9. The diagram below shows a sector of a circle with radius 6 cm and angle 0.9 radians.



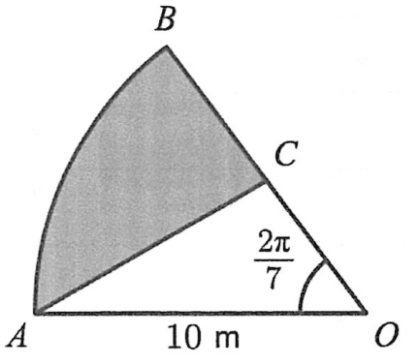
a) Find the area of the shaded segment.
b) Find the perimeter of the shaded segment.

2.01 — Question 10. The diagram below shows the circle with equation $x^2 + y^2 - 4x + 14y + 19 = 0$. It is given that OAB is a sector with angle 2.25 radians.

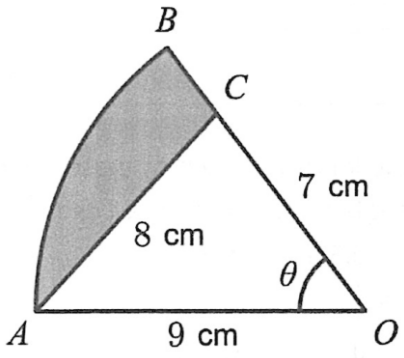


a) Find the area of the shaded segment.
b) What percentage of the circle is shaded in the diagram?

2.01 — Question 11. The diagram below shows a sector, OAB, of a circle with radius 10 m and angle $\frac{2\pi}{7}$ radians. It is given that C is the midpoint of O and B. Find the shaded area.



2.01 — Question 12. The diagram below shows a sector, OAB, of a circle with radius 9 cm and angle θ radians. It is given that AC has length 8 cm, and that OC has length 7 cm.

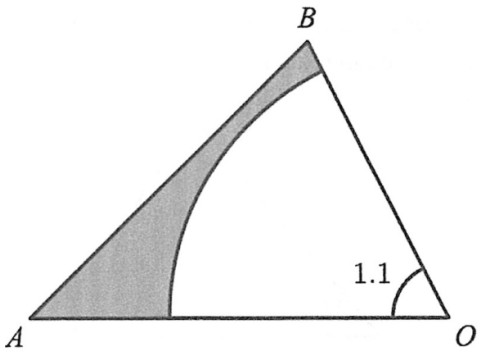


a) Show that $\cos\theta = \frac{11}{21}$

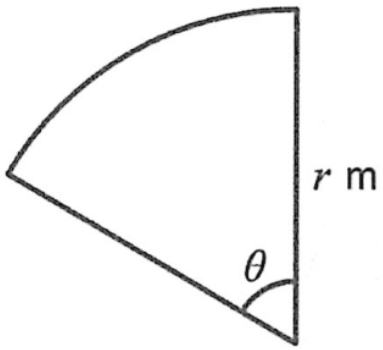
b) Hence show that $\sin\theta = \frac{8\sqrt{5}}{21}$

c) Find the area of the shaded region.

2.01 — Question 13. The diagram below shows the triangle OAB. Within the triangle is a sector of a circle, where angle AOB is 1.1 radians. It is given that $OA = 15$ cm and $OB = 5\sqrt{5}$ cm. The area of the shaded region is 20 cm^2. Find the perimeter of the shaded region to 3 significant figures.

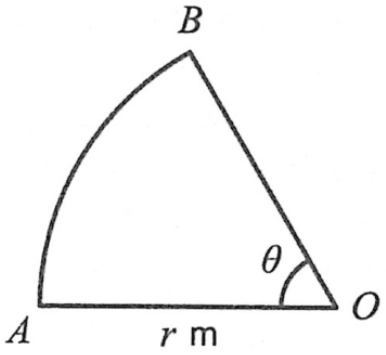


2.01 — Question 14. The diagram below shows a sector of a circle with radius r m and angle θ radians. It is given that the area of the sector is 2 m^2 and the perimeter is 6 m.

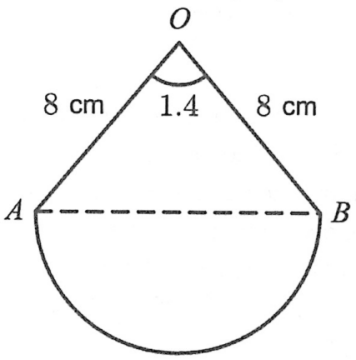


a) Show that $r^2 - 3r + 2 = 0$
b) Hence find the two possible values of θ

2.01 — Question 15. The diagram below shows a sector of a circle with radius r m and angle θ radians. It is given that the area of the sector is 45 m^2 and the perimeter is six times the length of the arc AB. Find the values of r and θ

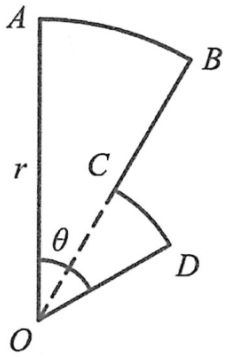


2.01 — Question 16. The diagram below shows the cross-section of a design for a new spinning toy.

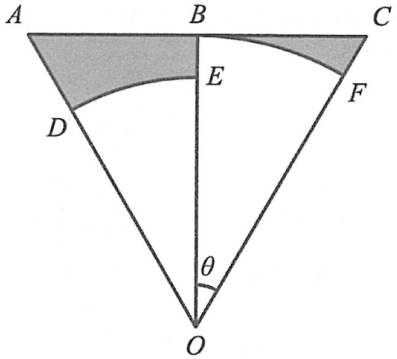


The shape is constructed from an isosceles triangle OAB and a semi-circle with diameter AB. The toy is redesigned so that the cross-section is instead a sector of a circle with the same area as the original design. The sector has an angle of 1.4 radians and radius r cm. Determine the value of r correct to 3 significant figures.

2.01 — Question 17. A company logo is made from two similar sectors, as shown in the diagram below, where OA is twice the length of OD. The length of OA is r and the angle AOB is θ radians. It is given that the perimeter and area of the whole logo is 9.2 and 2 respectively. Show that there is only one valid pair of values for r and θ

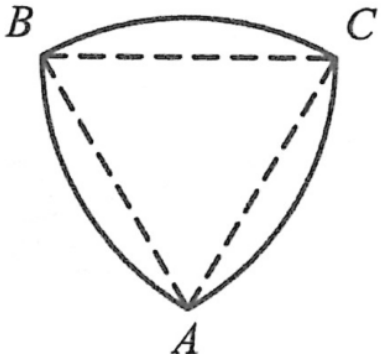


2.01 — Question 18. It is given that OAC is an equilateral triangle with side length r, and the angle BOC is θ radians. It is also given that OBF and ODE are sectors. B is the midpoint of AC.

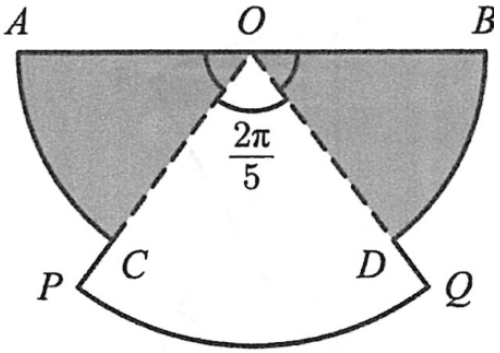


Given that $OB = 5BE$, show that the shaded area can be written in the form $(a\sqrt{3} + b\pi)r^2$, where a and b are rational numbers.

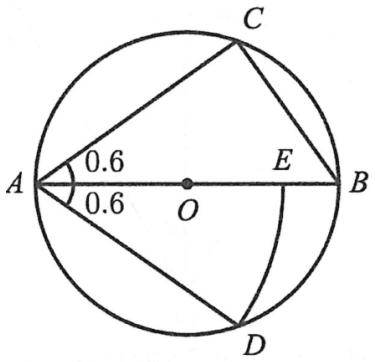
2.01 — Question 19. The diagram below shows an equilateral triangle, ABC, with side length 5 cm. Glued to the edge of each side of the triangle is the segment of a circle created at the opposite corner. Find the total exact area of the badge.



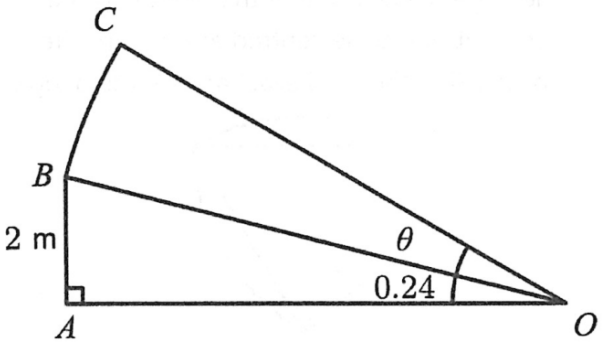
2.01 — Question 20. The diagram below shows part of a design for a playground. It is made from three sectors. The length of the arc PQ is 12π m. The two sectors OAC and OBD are precisely the same size, and their combined area is equal to the area of the sector OPQ. It is given that AOB is a straight line. Find the length of AB.



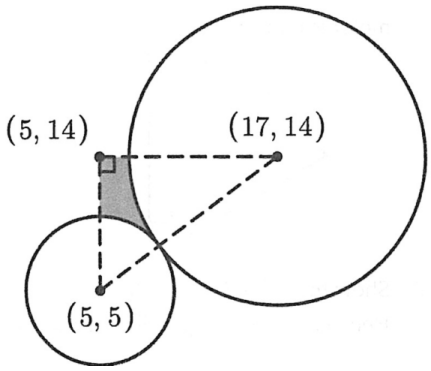
2.01 — Question 21. The diagram below shows the circle with equation $x^2 + y^2 - 6x + 12y - 4 = 0$. It is given that AB is a diameter of the circle, and both C and D lie on the circle. The angle CAB and the angle DAE are both equal to 0.6 radians. ADE is a sector of a circle centred at A. Find the area of the sector ADE to 3 significant figures.



2.01 — Question 22. In the diagram below, OAB is a right-angled triangle and OBC is a sector of a circle centred at O. Triangle OAB and sector OBC have the same area. The angle BOC is θ radians. Find the shortest distance from C to the line AO

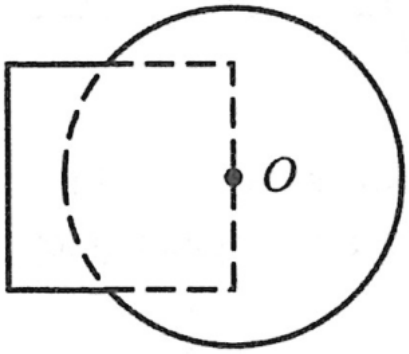


2.01 — Question 23. The diagram below shows the two circles with equations $(x-5)^2+(y-5)^2=25$ and $(x-17)^2+(y-14)^2=100$

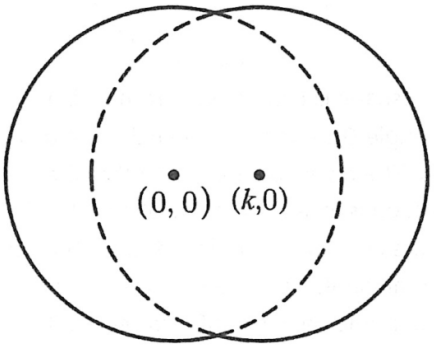


a) Show that the circles intersect at a single point.
b) Find the shaded area.

2.01 — Question 24. A square is placed over the top of a circle, as shown in the diagram below, to create a new shape. The circle, centred at O, has equation $x^2+y^2=36$. The point O is the midpoint of one of the sides of the square. The square has an area of 64 units^2. Find the perimeter of the new shape, ignoring any dashed lines.



2.01 — Question 25. The diagram below shows the two circles $x^2 + y^2 = 16$ and $(x - k)^2 + y^2 = 16$. It is given that the perimeter of the shape formed by the overlapping circles is 10π. Find the exact value of k, given that $k > 0$



2.02 — Question 1. Sketch the following graphs for $0 \leq x \leq 2\pi$:

a) $y = \sin x$

b) $y = \cos x$

c) $y = \tan x$

2.02 — Question 2. Solve the following equations:

a) $\sin\theta = 0, \quad 0 \leq \theta < 2\pi$

b) $\sin\theta = 1, \quad 0 \leq \theta < 2\pi$

c) $\sin\theta = -1, \quad 0 \leq \theta < 2\pi$

2.02 — Question 3. Solve the following equations:

a) $\cos\theta = 0, \quad 0 \leq \theta < 2\pi$

b) $\cos\theta = 1, \quad 0 \leq \theta < 2\pi$

c) $\cos\theta = -1, \quad 0 \leq \theta < 2\pi$

2.02 — Question 4. Solve the following equations:

a) $\tan\theta = 0, \quad 0 \leq \theta < 2\pi$

b) $\tan\theta = 1, \quad 0 \leq \theta < 2\pi$

c) $\tan\theta = -1, \quad 0 \leq \theta < 2\pi$

2.02 — Question 5. Find the values of x for which $y = \sin x$ is a decreasing function, for $0 \leq x < 2\pi$.

2.02 — Question 6. Find the values of x for which $y = \cos x$ is an increasing function, for $0 \leq x < 2\pi$.

2.02 — Question 7. Solve the following equations:

a) $\sin\theta = \frac{1}{4}, \quad 0 \leq \theta < 2\pi$

b) $\cos\theta = \frac{3}{4}, \quad 0 \leq \theta < 2\pi$

c) $\tan\theta = \frac{5}{4}, \quad 0 \leq \theta < 2\pi$

2.02 — Question 8. Solve the following equations:

a) $\sin\theta = 0.9, \quad -2\pi \leq \theta < 0$

b) $\cos\theta = \frac{1}{9}, \quad -2\pi \leq \theta < 0$

c) $\tan\theta = \frac{6}{5}, \quad -2\pi \leq \theta < 0$

2.02 — Question 9. Solve the following equations:

a) $\sin\theta = -\frac{1}{6}, \quad 0 \leq \theta < 2\pi$

b) $\cos\theta = -\frac{8}{9}, \quad 0 \leq \theta < 2\pi$

c) $\tan\theta = -\frac{1}{3}, \quad 0 \leq \theta < 2\pi$

2.02 — Question 10. Solve the following equations:

a) $\sin\theta = -0.66, \quad -2\pi \leq \theta < 0$

b) $\cos\theta = -0.55, \quad -2\pi \leq \theta < 0°$

c) $\tan\theta = -8.6, \quad -2\pi \leq \theta < 0$

2.02 — Question 11. Solve the following equations:

a) $\sin\theta = \frac{4}{9}, \quad -2\pi \leq \theta < 2\pi$

b) $\cos\theta = 0.11, \quad -2\pi \leq \theta < 2\pi$

c) $\tan\theta = 3.2, \quad -2\pi \leq \theta < 2\pi$

d) $\sin\theta = -0.83, \quad -2\pi \leq \theta < 2\pi$

e) $\cos\theta = -\frac{10}{99}, \quad -2\pi \leq \theta < 2\pi$

f) $\tan\theta = -6.6, \quad -2\pi \leq \theta < 2\pi$

2.02 — Question 12. Find the values of x for which $y = \sin x$ is an increasing function, for $-5\pi \leq x \leq 0$.

2.02 — Question 13. Find the values of x for which $y = \cos x$ is a decreasing function, for $7\pi \leq x \leq 10\pi$.

2.02 — Question 14. Solve the following equations:

a) $\sin\theta = \frac{3}{16}, \quad 0 \leq \theta < 4\pi$

b) $\cos\theta = -\frac{5}{11}, \quad 3\pi \leq \theta < 6\pi$

c) $\tan\theta = -0.94, \quad -7\pi \leq \theta < -5\pi$

2.02 — Question 15. Solve the following equations:

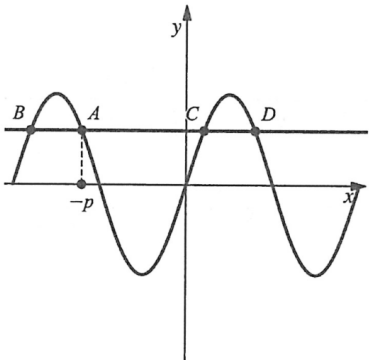
a) $\frac{1}{\sin\theta} = 16.5$ for $0 \leq \theta < \pi$

b) $\frac{14}{\tan\theta} = 19$ for $\pi \leq \theta < 4\pi$

c) $\frac{0.02}{\cos\theta} = 0.3$ for $-\frac{5\pi}{2} \leq \theta < -\frac{3\pi}{2}$

2.02 — Question 16. A student is solving a problem with a triangle. They correctly use the sine rule to find $\frac{\sin A}{150} = \frac{\sin 1^c}{135}$. Find the two possible values of A.

2.02 — Question 17. The graph below shows the curve $y = \sin x$ for $-2\pi \leq x \leq 2\pi$, and the line $y = k$ where $0 < k < 1$. The point A has x-coordinate $-p$, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.

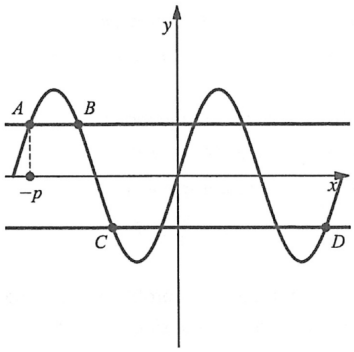


2.02 — Question 18. It is given that $\sin\frac{\pi}{6} = \frac{1}{2}$.
Explain why $\sin\frac{17\pi}{6} = \frac{1}{2}$.

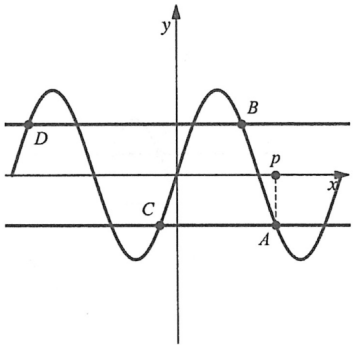
2.02 — Question 19. It is given that $\cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$.
Explain why $\cos(-\frac{29\pi}{4}) = -\frac{\sqrt{2}}{2}$.

2.02 — Question 20. It is given that $\tan(-\frac{39\pi}{4}) = 1$.
Explain why $\tan\frac{19\pi}{4} = -1$.

2.02 — Question 21. The graph below shows the curve $y = \sin x$ for $-2\pi \leq x \leq 2\pi$, and the lines $y = k$ and $y = -k$ where $0 < k < 1$. The point A has x-coordinate $-p$, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



2.02 — Question 22. The graph below shows the curve $y = \sin x$ for $-2\pi \leq x \leq 2\pi$, and the lines $y = k$ and $y = -k$ where $0 < k < 1$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



2.02 — Question 23. Solve the following equations:

a) $\sin^2\theta = \frac{1}{25}$ for $0 \leq \theta < 3\pi$

b) $16\tan^2\theta = 19$ for $-3\pi \leq \theta < -\pi$

c) $25\cos^2\theta = \frac{1}{4}$ for $\pi \leq \theta < 4\pi$

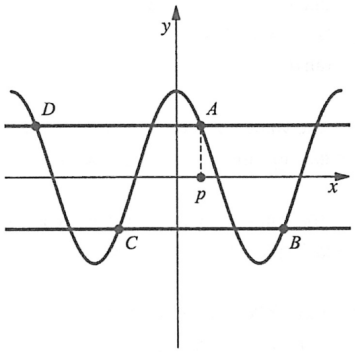
2.02 — Question 24. Solve the following equations:

a) $\sin^2\theta = 8\sin\theta$ for $0 \leq \theta < 2\pi$

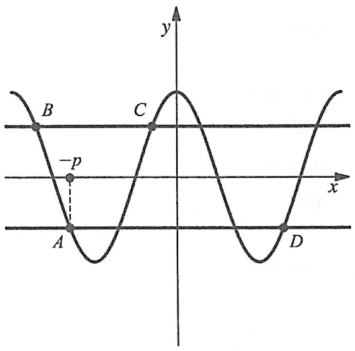
b) $9\tan^2\theta - 5\tan\theta = 0$ for $0 \leq \theta < 2\pi$

c) $-6\cos\theta = 11\cos^2\theta$ for $0 \leq \theta < 2\pi$

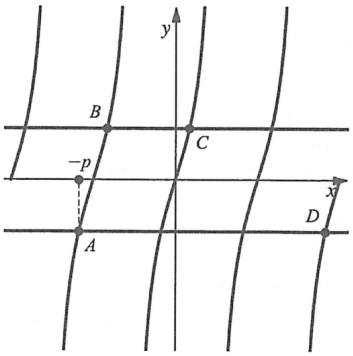
2.02 — Question 25. The graph below shows the curve $y = \cos x$ for $-2\pi \leq x \leq 2\pi$, and the lines $y = k$ and $y = -k$ where $0 < k < 1$. The point A has x-coordinate p, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



2.02 — Question 26. The graph below shows the curve $y = \cos x$ for $-2\pi \leq x \leq 2\pi$, and the lines $y = k$ and $y = -k$ where $0 < k < 1$. The point A has x-coordinate $-p$, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



2.02 — Question 27. The graph below shows the curve $y = \tan x$ for $-2\pi \leq x \leq 2\pi$, and the lines $y = k$ and $y = -k$ where $0 < k < 1$. The point A has x-coordinate $-p$, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



2.02 — Question 28. Solve the following equations:

a) $5\sin\theta\cos\theta = 3\cos\theta$ for $0 \leq \theta < 2\pi$

b) $\frac{2}{5}\sin\theta\tan\theta + 3\sin\theta = 0$ for $0 \leq \theta < 2\pi$

c) $29\sin\theta - 31\sin\theta\cos\theta = 0$ for $0 \leq \theta < 2\pi$

2.02 — Question 29. Solve the following equations:

a) $6\sin^2\theta - \sin\theta - 1 = 0$ for $0 \leq \theta < 2\pi$

b) $15\cos^2\theta - 2\cos\theta - 1 = 0$ for $0 \leq \theta < 2\pi$

c) $18\tan^2\theta + 15\tan\theta + 2 = 0$ for $0 \leq \theta < 2\pi$

2.02 — Question 30. Solve the following equations for $0 \leq \theta < 2\pi$:

a) $32\sin\theta = \frac{9}{\sin\theta} - 12$

b) $54\cos\theta = 51 + \frac{14}{\cos\theta}$

c) $20\tan\theta = 48 + \frac{77}{\tan\theta}$

2.02 — Question 31. It is given that $f(x) = 5x^2 + 70x + 219$.

a) Write $f(x)$ in the form $a(x+b)^2 + c$, where a, b and c are integers.

b) Write down the minimum point of the curve with equation $y = f(x)$

It is given that $g(\theta) = 5\cos^2\theta - 70\cos\theta + 219$.

c) Using your answers to a) and b), find the minimum value of $g(\theta)$ and the smallest positive value of θ, in radians, for which this minimum occurs.

2.02 — Question 32. Write down the period of the following functions in radians:

a) $y = \sin x$

b) $y = \cos x$

c) $y = \tan x$

d) $y = \sin 3x$

e) $y = \cos 2x$

f) $y = \tan 6x$

g) $y = \sin \frac{1}{5}x$

h) $y = \cos \frac{8}{3}x$

i) $y = \tan \frac{9}{16}x$

2.02 — Question 33. State the number of solutions to the following equations. You are not required to solve them.

a) $\sin 2x = 0.7$ for $0 \leq x < 2\pi$

b) $\sin 2x = 0.7$ for $0 \leq x < 3\pi$

c) $\sin 2x = 0.7$ for $-4\pi \leq x < 8\pi$

2.02 — Question 34. State the number of solutions to the following equations. You are not required to solve them.

a) $\cos 2x = 0.2$ for $-\pi \leq x < \pi$

b) $\cos 3x = 0.2$ for $0 \leq x < 5\pi$

c) $\cos 4x = 0.2$ for $-6\pi \leq x < \frac{3\pi}{2}$

2.02 — Question 35. State the number of solutions to the following equations. You are not required to solve them.

a) $\tan 2x = 2.1$ for $-2\pi \leq x < 3\pi$

b) $\tan 3x = 2.1$ for $-5\pi \leq x < 10\pi$

c) $\tan 4x = 2.1$ for $-25\pi \leq x < 40\pi$

2.02 — Question 36. Write down the period of the following functions in radians:

a) $y = \sin(x + \pi)$

b) $y = \cos(x - \frac{2\pi}{3})$

c) $y = \tan(x - 0.1)$

d) $y = \sin(4\pi - x)$

e) $y = \cos(0.9 - x)$

f) $y = \tan(1.3 - x)$

g) $y = \sin(3x + \frac{\pi}{9})$

h) $y = \cos(6x - 0.34)$

i) $y = \tan(12x + \frac{2\pi}{5})$

j) $y = \sin(0.4 - \frac{1}{6}x)$

k) $y = \cos(\frac{0.56 - x}{0.16})$

l) $y = \tan(\frac{5\pi - 15x}{12})$

2.02 — Question 37. Solve the following equations:

a) $\sin(\theta + 0.32) = 0.45$ for $0 \leq \theta < 2\pi$

b) $\sin(\theta - \frac{8\pi}{9}) = \frac{6}{7}$ for $0 \leq \theta < 2\pi$

c) $\sin(\theta - 1.2) = -0.33$ for $0 \leq \theta < 2\pi$

2.02 — Question 38. Solve the following equations:

a) $\cos(\theta - 0.97) = 0.2$ for $0 \leq \theta < 2\pi$

b) $\cos(\theta + 2.3) = 0.49$ for $0 \leq \theta < 2\pi$

c) $\cos(\theta - \frac{5\pi}{3}) = -0.89$ for $0 \leq \theta < 2\pi$

2.02 — Question 39. Solve the following equations:

a) $\tan(\theta + 2.01) = 6$ for $0 \leq \theta < 2\pi$

b) $\tan(\theta - \frac{\pi}{5}) = \frac{3}{8}$ for $0 \leq \theta < 2\pi$

c) $\tan(\theta + 0.99) = -8.2$ for $0 \leq \theta < 2\pi$

2.02 — Question 40. Find the values of x for which $y = \sin(x - \frac{\pi}{4})$ is a decreasing function, for $0 \leq x \leq 2\pi$.

2.02 — Question 41. Solve the following equations:

a) $\sin 2\theta = \frac{\sqrt{2}}{2}$ for $0 \leq \theta < 2\pi$

b) $\cos 2\theta = \frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$

c) $\tan 2\theta = \sqrt{3}$ for $0 \leq \theta < 2\pi$

2.02 — Question 42. Solve the following equations:

a) $\sin 3\theta = 0.76$ for $0 \leq \theta < 2\pi$

b) $\cos 3\theta = -0.95$ for $0 \leq \theta < 2\pi$

c) $\tan 3\theta = -0.32$ for $0 \leq \theta < 2\pi$

2.02 — Question 43. Solve the following equations:

a) $\sin \frac{1}{2}\theta = -0.28$ for $0 \leq \theta < 4\pi$

b) $\cos \frac{1}{5}\theta = \frac{2}{9}$ for $0 \leq \theta < 10\pi$

c) $\tan \frac{7}{5}\theta = -2.4$ for $0 \leq \theta < \pi$

2.02 — Question 44. Solve the following equations:

a) $\sin(3\theta - 0.31) = \frac{2}{7}$ for $0 \leq \theta < \pi$

b) $\cos(4\theta + 0.97) = \frac{6}{11}$ for $0 \leq \theta < \pi$

c) $\tan(5\theta + 1.45) = \frac{7}{8}$ for $0 \leq \theta < \frac{\pi}{2}$

2.02 — Question 45. Solve the following equations:

a) $\sin(5\theta + 3.2) = -0.16$ for $-\pi \leq \theta < 0$

b) $\cos(7\theta - 4.2) = -0.12$ for $\pi \leq \theta < \frac{3\pi}{2}$

c) $\tan(3\theta - 6.4) = -12$ for $2\pi \leq \theta < 4\pi$

2.02 — Question 46. Find the values of x for which $y = \cos 3x$ is a decreasing function, for $0 \leq x \leq \pi$.

2.02 — Question 47. Find the values of x for which $y = \sin \frac{1}{4}x$ is an increasing function, for $0 \leq x \leq 10\pi$.

2.02 — Question 48. Solve the following equations:

a) $\sin(\frac{\pi}{8} - 2\theta) = \frac{\sqrt{2}}{2}$ for $0 \leq \theta < 2\pi$

b) $\cos(\frac{3\pi}{5} - 3\theta) = \frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$

c) $\tan(\frac{3\pi}{10} - 4\theta) = 1$ for $0 \leq \theta < 2\pi$

2.02 — Question 49. Solve the following equations:

a) $\sin(1.2 - 4\theta) = -0.07$ for $0 \leq \theta < 2$

b) $\cos(2.1 - 5\theta) = 0.93$ for $0 \leq \theta < 1.5$

c) $\tan(4.3 - 7\theta) = -0.16$ for $0 \leq \theta < 1$

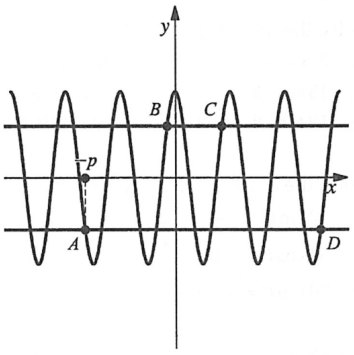
2.02 — Question 50. Solve the following equations:

a) $\sin^2 4\theta - \frac{1}{25} = 0$ for $-1.4 \leq \theta < 1.4$

b) $\cos^2 3\theta - \frac{1}{81} = 0$ for $-2 \leq \theta < 2$

c) $\tan^2 9\theta - 169 = 0$ for $-0.5 \leq \theta < 0.5$

2.02 — Question 51. The graph below shows the curve $y = \cos 3x$ for $-2\pi \leq x \leq 2\pi$, and the lines $y = k$ and $y = -k$ where $0 < k < 1$. The point A has x-coordinate $-p$, where $p > 0$. Write down the x-coordinates of B, C and D in terms of p.



2.02 — Question 52. Solve the following equations:

a) $\sin(\theta - \frac{\pi}{3}) = \sin\frac{\pi}{6}$ for $0 \leq \theta < 2\pi$

b) $\sin(\theta + \frac{2\pi}{5}) = \sin\frac{11\pi}{10}$ for $0 \leq \theta < 2\pi$

c) $\cos(\theta + \frac{3\pi}{7}) = \cos\frac{4\pi}{3}$ for $0 \leq \theta < 2\pi$

d) $\tan(\theta - \frac{5\pi}{8}) = \tan\frac{\pi}{16}$ for $0 \leq \theta < 2\pi$

2.02 — Question 53. Solve the following equations:

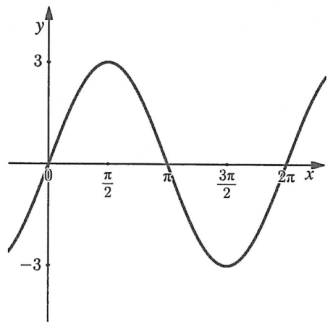
a) $\sin(\frac{\pi}{3} - \theta) = \sin\frac{46\pi}{9}$ for $0 \leq \theta < 2\pi$

b) $\cos(2\theta - \frac{5\pi}{6}) = \cos\frac{11\pi}{12}$ for $0 \leq \theta < 2\pi$

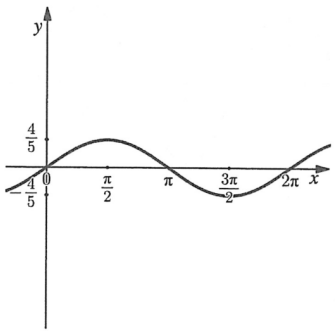
c) $\tan(\frac{7\pi}{6} - 4\theta) = \tan\frac{8\pi}{3}$ for $0 \leq \theta < 2\pi$

2.03 — Question 1. Find the equation of each of the following graphs in the form $y = a\sin x$

a)

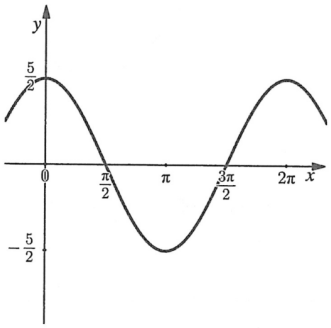


b)

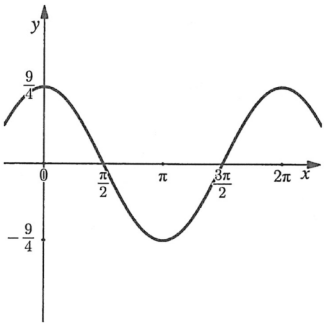


2.03 — Question 2. Find the equation of each of the following graphs in the form $y = a\cos x$

a)



b)



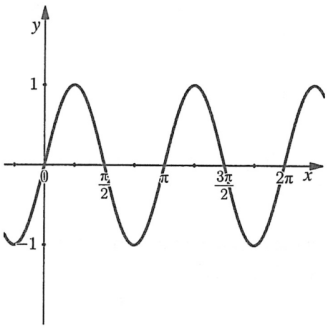
2.03 — Question 3. Sketch the following graphs for $0 \leq x \leq 2\pi$:

a) $y = \frac{1}{3}\sin x$

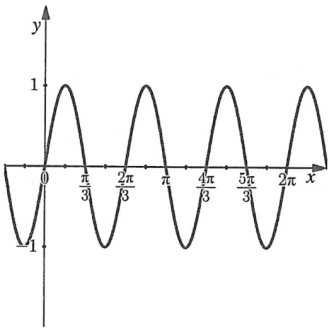
b) $y = \frac{8}{7}\cos x$

2.03 — Question 4. Find the equation of each of the following graphs in the form $y = \sin ax$

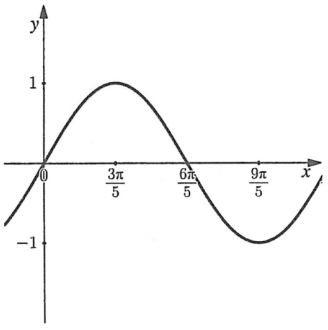
a)



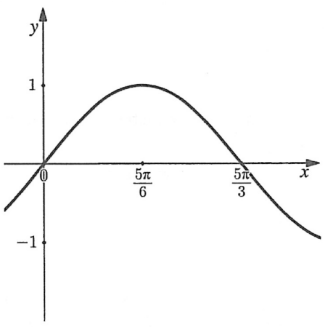
b)



c)



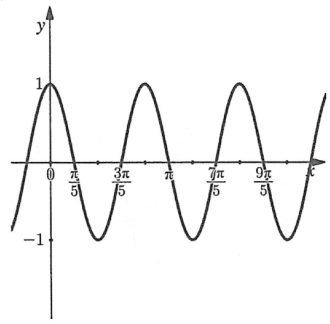
d)



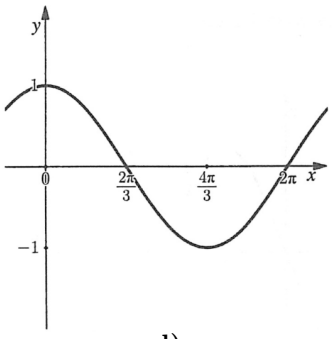
2.03 — Question 5. Sketch the graph of $y = -4\sin x$ for $0 \leq x \leq 2\pi$

2.03 — Question 6. Find the equation of each of the following graphs in the form $y = \cos ax$

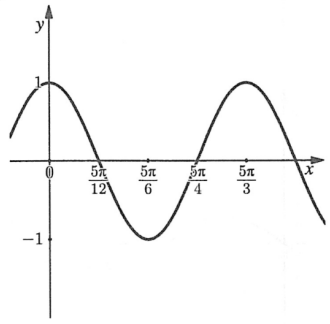
a)



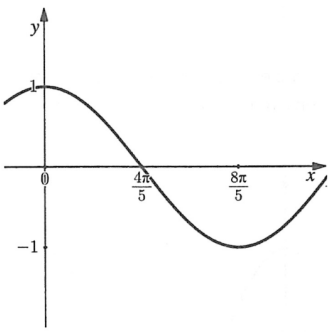
b)



c)



d)

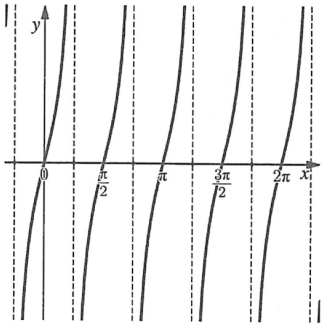


2.03 — Question 7. Sketch the graph of $y = -\frac{3}{2}\cos x$ for $0 \leq x \leq 2\pi$

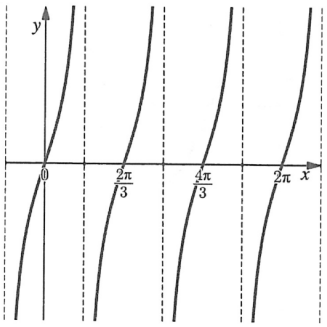
2.03 — Question 8. Sketch the graph of $y = \sin(-\frac{2}{3}x)$ for $0 \leq x \leq 2\pi$

2.03 — Question 9. Find the equation of each of the following graphs in the form $y = \tan ax$

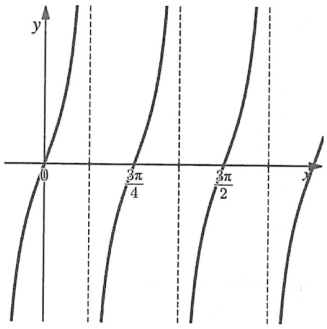
a)



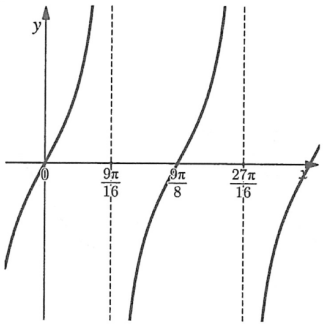
b)



c)



d)



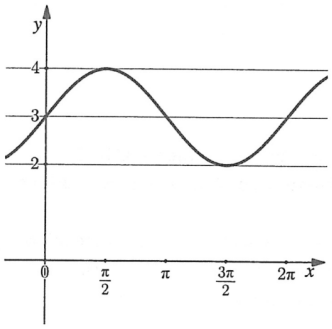
2.03 — Question 10. Sketch the following graphs for $0° \leq x \leq 2\pi$:

a) $y = \tan \frac{2}{5}x$

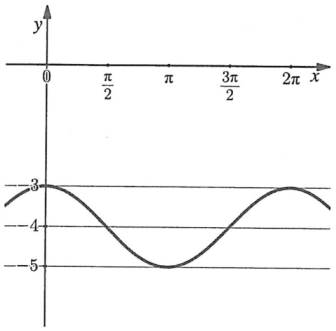
b) $y = \tan \frac{10}{9}x$

2.03 — Question 11. Find the equation of each of the following graphs in either the form $y = a + \sin x$ or the form $y = a + \cos x$

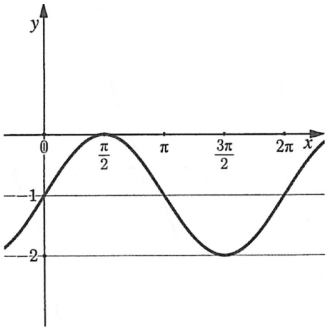
a)



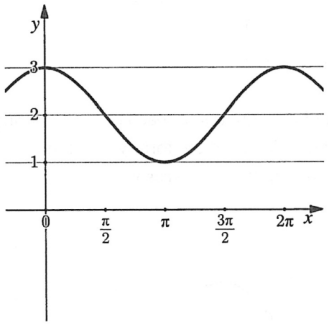
b)



c)

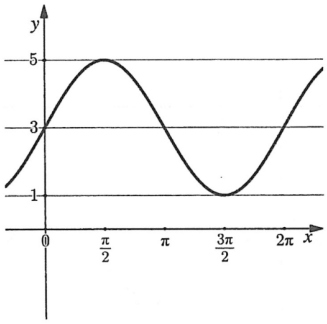


d)

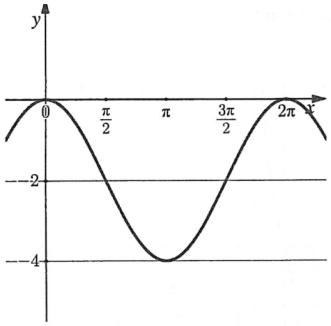


2.03 — Question 12. Prerequisite knowledge for Q12–Q13: Combinations of Graph Transformations
Find the equation of each of the following graphs in the form $y = a + b\sin x$ or the form $y = a + b\cos x$

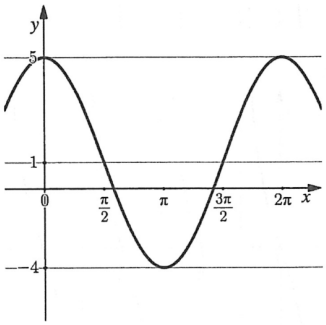
a)



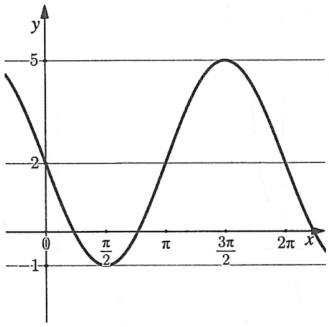
b)



c)



d)



2.03 — Question 13. Sketch the following graphs for $0 \leq x \leq 2\pi$:

a) $y = 1 + 3\sin x$

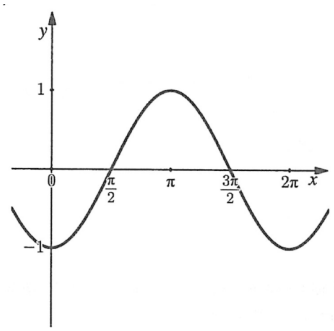
b) $y = -2 + 3\cos x$

c) $y = \frac{3}{2} - 4\sin x$

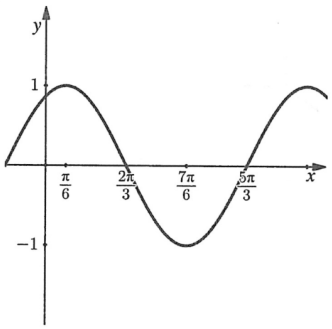
d) $y = -\frac{2}{3} - 2\cos x$

2.03 — Question 14. Find the equation of each of the following graphs in the form $y = \sin(x+a)$ where $-\pi < a < \pi$

a)

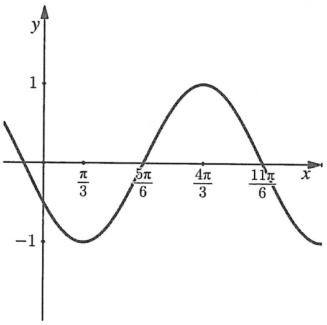


b)

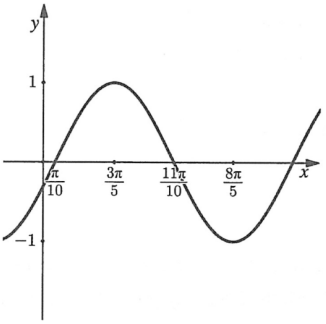


2.03 — Question 15. Find the equation of each of the following graphs in the form $y = \cos(x+a)$ where $-\pi < a < \pi$

a)

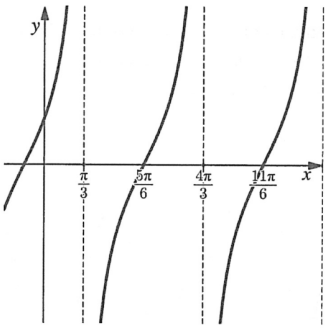


b)

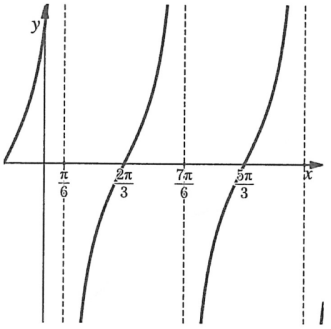


2.03 — Question 16. Find the equation of each of the following graphs in the form $y = \tan(x + a)$ where $-\frac{\pi}{2} < a < \frac{\pi}{2}$

a)



b)



2.03 — Question 17. It is given that k is a constant where $k > 1$. Sketch the following graphs for $0 \leq x \leq 2\pi$:

a) $y = -k + \sin x$

b) $y = k + \cos x$

2.03 — Question 18. Sketch the following graphs for $0 \leq x \leq 2\pi$:

a) $y = \sin(x - \frac{\pi}{3})$

b) $y = \cos(x + \frac{\pi}{3})$

c) $y = \tan(x - \frac{\pi}{5})$

d) $y = \sin(x + \frac{4\pi}{5})$

e) $y = \cos(x - \frac{\pi}{6})$

f) $y = \tan(x + \frac{3\pi}{8})$

2.03 — Question 19. It is given that k is a constant where $0 < k < \frac{\pi}{2}$. Sketch the following graphs for $0 \leq x \leq 2\pi$:

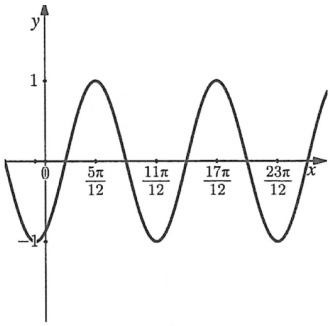
a) $y = \sin(x - k)$

b) $y = \tan(x + k)$

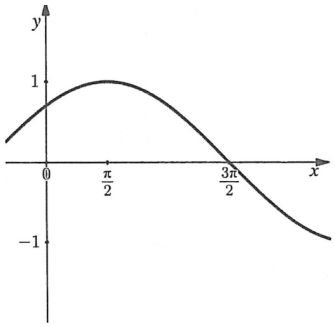
2.03 — Question 20. Prerequisite knowledge for Q20–Q25: Combinations of Graph Transformations

Find the equation of each of the following graphs in the form $y = \sin(ax + b)$ where $-\pi < b < \pi$

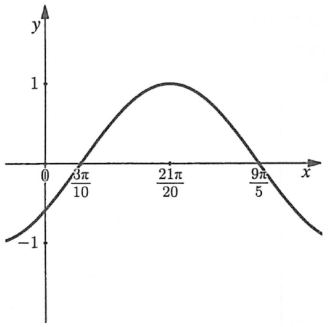
a)



b)

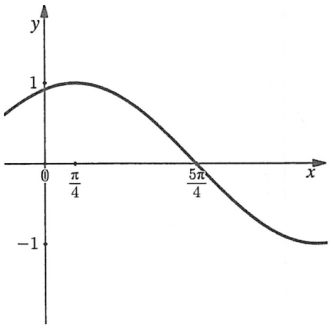


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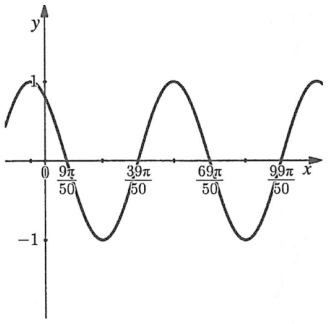


2.03 — Question 21. Find the equation of each of the following graphs in the form $y = \cos(ax + b)$ where $-\pi < b < \pi$

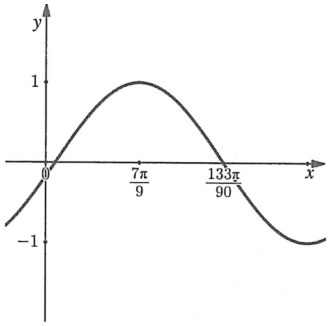
a)



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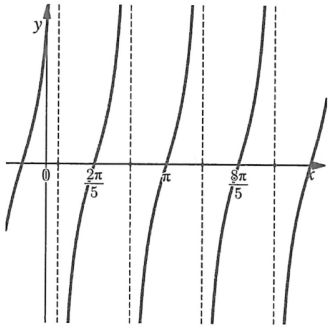


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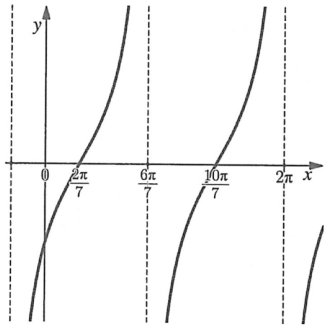


2.03 — Question 22. Find the equation of each of the following graphs in the form $y = \tan(ax + b)$ where $-\frac{\pi}{2} < b < \frac{\pi}{2}$

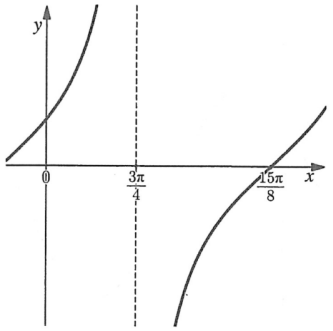
a)



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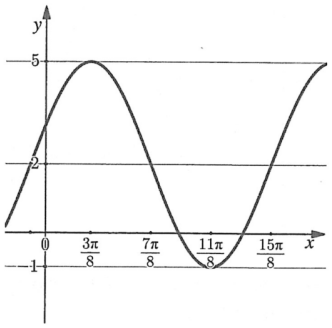


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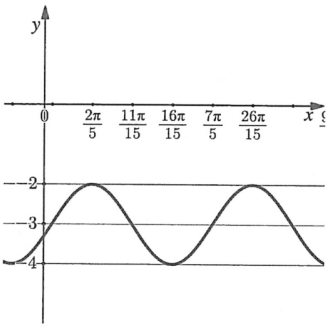


2.03 — Question 23. Find the equation of each of the following graphs in the form $y = a + b\sin(cx + d)$ where $-\pi < d < \pi$

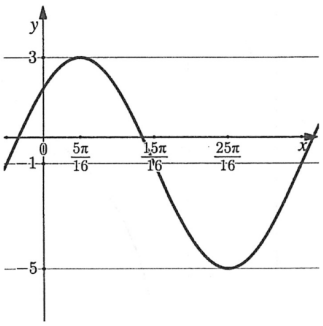
a)



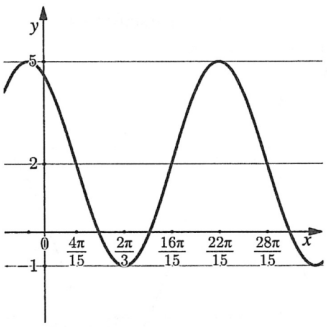
b)



c)

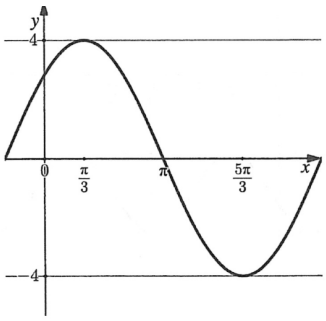


d)

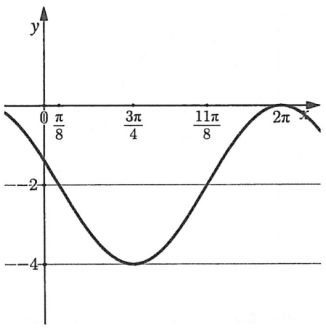


2.03 — Question 24. Find the equation of each of the following graphs in the form $y = a + b\cos(cx + d)$ where $-\pi < c < \pi$

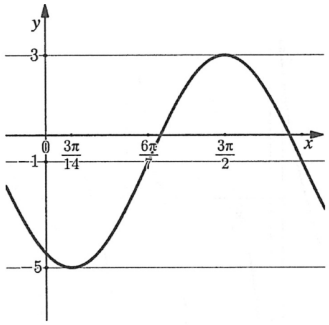
a)



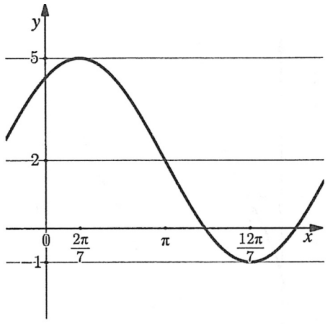
b)



c)



d)



2.03 — Question 25. It is given that a and b are positive constants. Sketch the following graphs for $0 \leq x < 2\pi$:

a) $y = a + b\cos x$ where $a > b$

b) $y = a + b\cos x$ where $a = b$

c) $y = a + b\cos x$ where $a < b$

2.04 — Question 1. It is given that $\sin^2 x = \frac{4}{5}$ and x is acute. Find the exact values of:

a) $\cos x$

b) $\tan x$

2.04 — Question 2. It is given that $\cos^2 x = \frac{8}{9}$ and x is acute. Find the exact values of:

a) $\sin x$

b) $\tan x$

2.04 — Question 3. It is given that $\tan^2 x = \frac{25}{6}$ and x is acute. Find the exact values of:

a) $\sin x$

b) $\cos x$

2.04 — Question 4. It is given that $\sin^2 x = \frac{5}{12}$ and x is obtuse. Find the exact values of:

a) $\cos x$

b) $\tan x$

2.04 — Question 5. It is given that $\cos^2 x = \frac{12}{13}$ and x is obtuse. Find the exact values of:

a) $\sin x$

b) $\tan x$

2.04 — Question 6. It is given that $\tan^2 x = \frac{30}{7}$ and x is obtuse. Find the exact values of:

a) $\sin x$

b) $\cos x$

2.04 — Question 7. It is given that $\sin^2 x = \frac{3}{16}$ and $\pi < x < \frac{3\pi}{2}$. Find the exact values of:

a) $\cos x$

b) $\tan x$

2.04 — Question 8. It is given that $\cos^2 x = \frac{11}{12}$ and $\pi < x < \frac{3\pi}{2}$. Find the exact values of:

a) $\sin x$

b) $\tan x$

2.04 — Question 9. It is given that $\tan^2 x = \frac{18}{11}$ and $\pi < x < \frac{3\pi}{2}$. Find the exact values of:

a) $\sin x$

b) $\cos x$

2.04 — Question 10. It is given that $\sin^2 x = \frac{24}{25}$ and $\frac{3\pi}{2} < x < 2\pi$. Find the exact values of:

a) $\cos x$

b) $\tan x$

2.04 — Question 11. It is given that $\cos^2 x = \frac{24}{49}$ and $\frac{3\pi}{2} < x < 2\pi$. Find the exact values of:

a) $\sin x$

b) $\tan x$

2.04 — Question 12. It is given that $\tan^2 x = \frac{60}{17}$ and $\frac{3\pi}{2} < x < 2\pi$. Find the exact values of:

a) $\sin x$

b) $\cos x$

2.04 — Question 13. It is given that $\cos \frac{\pi}{10} = \sqrt{\frac{5}{8} + \frac{\sqrt{5}}{8}}$

a) Find the exact value of $\sin \frac{\pi}{10}$

b) Using your answer to a), show that $\tan \frac{\pi}{10} = \sqrt{1 - \frac{2}{\sqrt{5}}}$

2.04 — Question 14. Serkan is attempting to solve the equation $3\sin x = \tan x$ for $0° \leq x < 360°$. His attempt is shown below:

$$\begin{aligned} 3\sin x &= \tan x \\ \Rightarrow 3\sin x &= \frac{\sin x}{\cos x} \\ \Rightarrow 3\sin x\cos x &= \sin x \\ \Rightarrow 3\cos x &= 1 \\ \Rightarrow \cos x &= \frac{1}{3} \\ \Rightarrow x &= 1.23 \text{ or } 5.05 \text{ to 3sf} \end{aligned}$$

Explain the two errors that Serkan has made.

2.04 — Question 15. Solve the following equations:

a) $\sin\theta = \cos\theta,\ 0 \leq \theta < 4\pi$

b) $20\sin\theta + 3\cos\theta = 0,\ 0 \leq \theta < 2\pi$

c) $11\cos\theta - 4\sin\theta = 0,\ 0 \leq \theta < 2\pi$

2.04 — Question 16. Solve the following equations:

a) $\sin 2\theta = \cos 2\theta,\ 0 \leq \theta < 3\pi$

b) $7\sin 4\theta = 2\cos 4\theta,\ 0 \leq \theta < \frac{\pi}{2}$

c) $10\sin \frac{1}{2}\theta + 3\cos \frac{1}{2}\theta = 0,\ 0 \leq \theta < 5\pi$

d) $19\cos \frac{1}{4}\theta - 16\sin \frac{1}{4}\theta = 0,\ 0 \leq \theta < 20\pi$

2.04 — Question 17. Solve the following equations:

a) $6\tan\theta + \cos\theta = 0,\ 0 \leq \theta < 2\pi$

b) $\sin\theta\tan\theta + 16 = 0,\ 0 \leq \theta < 2\pi$

c) $12\tan \frac{1}{2}\theta + \cos \frac{1}{2}\theta = 0,\ 0 \leq \theta < 8\pi$

2.04 — Question 18. Solve the following equations:

a) $3\cos^2\theta = 8\sin\theta,\ 0 \leq \theta < \pi$

b) $9\sin^2\theta = 80\cos\theta,\ 0 \leq \theta < 3\pi$

c) $12\cos^2\theta = 143\sin\theta,\ 0 \leq \theta < 2\pi$

d) $50\sin^2\theta = 2499\cos\theta,\ -\pi \leq \theta < \pi$

2.04 — Question 19. Solve the following equations:

a) $4\cos^2\theta + 9\sin\theta - 6 = 0,\ 0 \leq \theta < 2\pi$

b) $9\sin^2\theta + 3\cos\theta - 7 = 0,\ 0 \leq \theta < 2\pi$

c) $15\cos^2\theta - 31\sin\theta - 25 = 0,\ 0 \leq \theta < 2\pi$

d) $27\sin^2\theta + 12\cos\theta - 7 = 0,\ 0 \leq \theta < 2\pi$

2.04 — Question 20. Solve the following equations:

a) $8\sin^2\theta - 26\sin\theta - 16\cos^2\theta + 21 = 0,\ 0 \leq \theta < 2\pi$

b) $13\cos^2\theta + 13\cos\theta - 22\sin^2\theta + 10 = 0,\ 0 \leq \theta < 2\pi$

c) $20\sin^2\theta - 4\cos^2\theta + 38\sin\theta - 51 = 0,\ 0 \leq \theta < 2\pi$

d) $50\sin^2\theta + 28 = 30\cos^2\theta + 31\cos\theta,\ 0 \leq \theta < 2\pi$

2.04 — Question 21.

a) Solve the equation $20\sin^2 x - 14\sin x - 1 = 4\cos^2 x$ for $-2\pi \leq x < 2\pi$

b) Hence find the smallest positive solution to the equation $20\sin^2(3\theta + \frac{\pi}{3}) - 14\sin(3\theta + \frac{\pi}{3}) - 1 = 4\cos^2(3\theta + \frac{\pi}{3})$

2.04 — Question 22. Solve the following equations:

a) $36\sin\theta + \frac{10}{\cos\theta} = 24\tan\theta + 15,\ 0 \leq \theta < 2\pi$

b) $3\sin\theta\tan\theta = 21\cos\theta + 14,\ 0 \leq \theta < 2\pi$

c) $\frac{6\cos\theta}{\tan\theta} = 29\sin\theta - 1,\ 0 \leq \theta < 2\pi$

d) $10\sin\theta\cos\theta - 2\cos^2\theta + 5\cos\theta = 2\sin\theta(2 + \sin\theta),\ 0 \leq \theta < 2\pi$

2.04 — Question 23. Solve the following equations:

a) $3\sin\theta\tan\theta = 2\sin\theta + \cos\theta,\ 0 \leq \theta < 2\pi$

b) $8\sin\theta\tan\theta + 39\sin\theta = 5\cos\theta,\ 0 \leq \theta < 2\pi$

c) $74\cos\theta\tan\theta = 45\sin\theta\tan\theta + 24\cos\theta,\ 0 \leq \theta < 2\pi$

d) $40\sin\theta\tan\theta = 15\cos\theta + 38\cos\theta\tan\theta,\ 0 \leq \theta < 2\pi$

2.04 — Question 24.

a) Solve the equation $18\sin^2 x + 3\cos x = 8$, $0 \leq x < 4\pi$

b) Hence solve the equation $18\sin^2 \frac{1}{3}\theta + 3\cos \frac{1}{3}\theta = 8$, $0 \leq \theta < 12\pi$

2.04 — Question 25.

a) Show that $\dfrac{2\sin x - 5\cos^2 x - 2}{5\sin x + 7} \equiv \sin x - 1$

b) Hence solve the equation $\dfrac{2\sin(\theta + \frac{\pi}{3}) - 5\cos^2(\theta + \frac{\pi}{3}) - 2}{5\sin(\theta + \frac{\pi}{3}) + 7} = 10\cos(\theta + \frac{\pi}{3}) - 1$ for $0 \leq \theta < 2\pi$

2.04 — Question 26.

a) Show that $\dfrac{25 - 40\cos^2 x - 38\sin x}{4\sin x - 5} \equiv 3 + 10\sin x$

b) Hence solve the equation $\dfrac{25 - 40\cos^2 2\theta - 38\sin 2\theta}{4\sin 2\theta - 5} = 3 - 8\cos 2\theta$ for $0 \leq \theta < \pi$

2.04 — Question 27. Solve the equation $\sqrt{18\sin^2\theta - 21\sin\theta\cos\theta} = 2\cos\theta$ for $0 \leq \theta < 2\pi$.

2.04 — Question 28. Prove the following trigonometric identities:

a) $\frac{\cos^2\theta-\cos^4\theta}{2\cos^2\theta-1} \equiv \frac{\sin\theta\cos\theta}{\frac{1}{\tan\theta}-\tan\theta}$

b) $\frac{\sin^4\theta-\sin^2\theta}{\sin^3\theta+2\sin^2\theta-\sin\theta-1} \equiv \frac{\sin\theta\cos\theta}{\frac{1}{\tan\theta}-\tan\theta+\cos\theta}$

2.05 — Question 1. State the period of each of the following functions:

a) $f(x) = \sin(x)$

b) $f(x) = \cos(4\pi x)$

c) $f(x) = \tan(\pi x)$

d) $f(x) = \sin(\frac{\pi}{2}x)$

e) $f(x) = \cos(\frac{\pi}{4}x)$

f) $f(x) = \tan(\frac{\pi}{2}x - \frac{\pi}{4})$

g) $f(x) = \sin(\frac{2\pi}{52}x)$

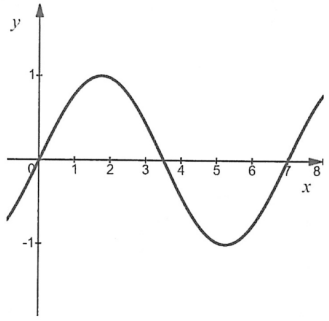
h) $f(x) = \cos(\frac{2\pi}{7}x + \frac{\pi}{18})$

i) $f(x) = \sin(\frac{2\pi}{365}(x - 30))$

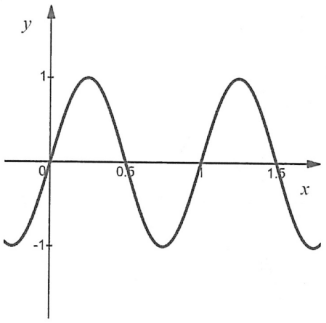
j) $f(x) = \cos(\frac{\pi}{26}(x + 7))$

2.05 — Question 2. In each of the following cases, the curve $y = \sin x$ has been stretched parallel to the x-axis. State the equation of each curve.

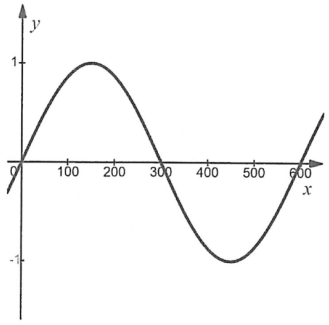
a)



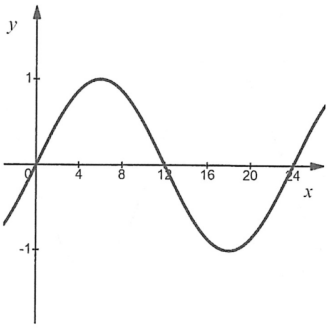
b)



c)

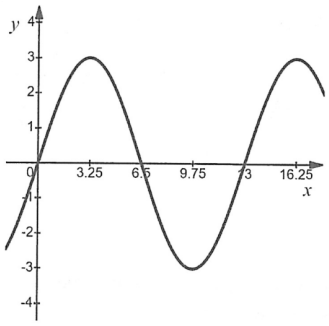


d)

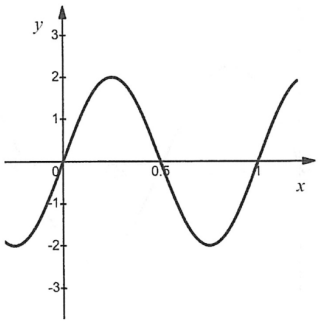


2.05 — Question 3. All the following graphs are transformations of $y = \sin x$
State the equation of each:

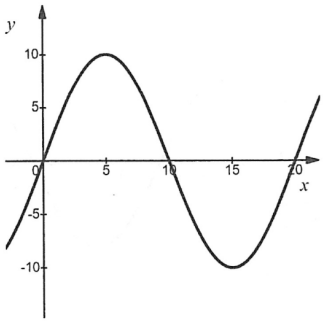
a)



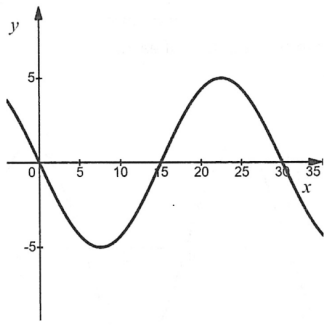
b)



c)



d)



2.05 — Question 4. State the amplitude of each of the following periodic functions:

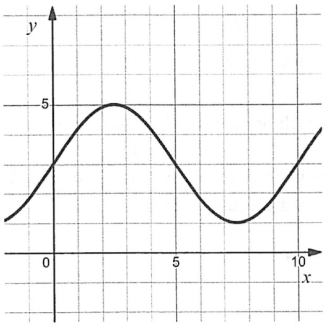
a) $y = \sin(3\pi x)$
b) $y = 4 + \cos(\pi x + 0.1)$
c) $y = 55 + 12\sin(\pi x)$
d) $y = 1 - 2\sin(x + \frac{\pi}{2})$

2.05 — Question 5. For each of the following functions state the minimum and maximum of the function:

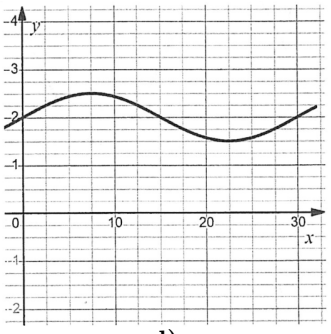
a) $y = \sin(2\pi x)$
b) $y = 3 + \cos(\pi x - \frac{\pi}{2})$
c) $y = 14 + 6\sin(30\pi x)$
d) $y = 2 - 7\cos(\frac{\pi}{2}x + 0.2)$

2.05 — Question 6. All the following graphs are all the form $y = a + b\sin(cx)$
In each case, find the equation of the graph:

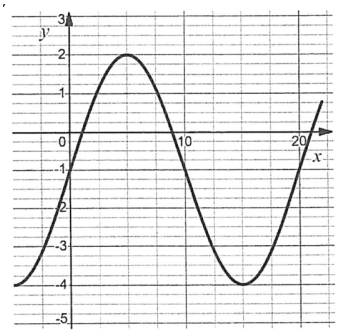
a)



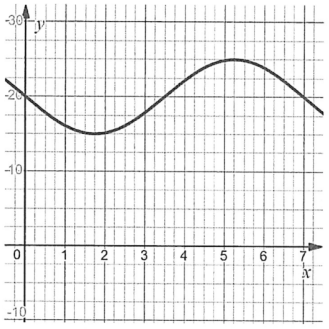
b)



c)



d)



2.05 — Question 7. The river Hamble is tidal. On one day the depth of the Hamble oscillates between a maximum of 3.6 metres at 8:00 am and a minimum of 2.0 metres at 10:30 pm. Kaelen decides to model the depth of the river, d metres by the equation

$$d = a + b\cos\left(\frac{2}{5}\pi t\right)$$

where t is the number of hours since 8:00 am.

a) State the values of a and b that Kaelen should use.
b) State the period of this equation.
c) Calculate the total time, according to this model, that the Hamble is less than 2.5 metres in depth within the first 7 hours.

2.05 — Question 8. The temperature, $T°\text{C}$ of an average human can be modelled by the equation

$$T = 36.8 - 0.64\sin\left(\frac{\pi}{12}(h+2)\right)$$

where h is the number of hours since midnight.

a) Find the maximum temperature of an average human.

b) Find the first time at which this maximum occurs after midnight.

The temperature, $T°\text{C}$ of an average dog varies between $38.3°\text{C}$ and $39.2°\text{C}$.

c) Find an equation to model a dog's temperature of the form $T = a - b\sin\left(\frac{\pi}{12}(h+2)\right)$

2.05 — Question 9. Prerequisite Knowledge for Q9: Graph Transformations (combinations) The time of sunset, t hours for the first day of the week x for a year may be modelled by the equation

$$t = 18.5 + 2.4\sin\left(\frac{2\pi}{52}(x+26)\right)$$

for $0 \leq x \leq 52$

a) Describe the transformations which map

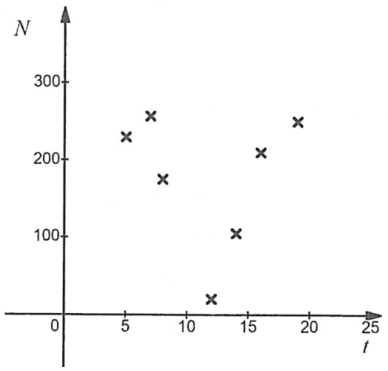
$$t = \sin\left(\frac{2\pi}{52}(x+26)\right)$$

onto

$$t = 18.5 + 2.4\sin\left(\frac{2\pi}{52}(x+26)\right)$$

b) Find the time of the sunset in the 14th week of the year.

2.05 — Question 10. The number of passengers waiting on a platform for a train is recorded over the course of a day, and they are plotted on a graph. Passengers are only counted once as they embark on a train before the next count of passengers takes place.



It is believed that the data can be modelled with the equation

$$N = 140 + 120\sin\left(\frac{\pi}{6}(t-3)\right)$$

where N is the number of passengers on the platform and t is the number of hours after midnight. According to the model, find

a) the difference in the number of passengers between 5 am and 9 am
b) the times at which the total number of passengers on the platform is over 200 over the 24-hour period.
c) Explain why it may not be suitable to use this function to model the data.

2.14 — Question 1. Sketch the following graphs, in radians, identifying the coordinates of any endpoints and the equations of any asymptotes.

a) $y = \arcsin x$

b) $y = \arccos x$

c) $y = \arctan x$

2.14 — Question 2. Sketch the following graphs, in radians, identifying the coordinates of any endpoints.

a) $y = 3\arcsin x$

b) $y = \frac{1}{4}\arcsin x$

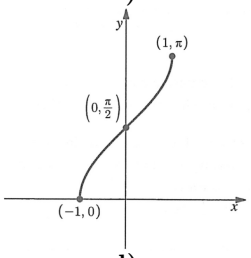
2.14 — Question 3. Sketch the following graphs, in radians, identifying the coordinates of any endpoints.

a) $y = \arcsin 3x$

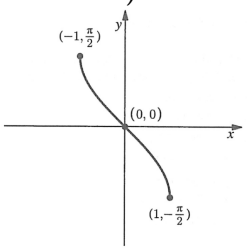
b) $y = \arcsin \frac{2}{5}x$

2.14 — Question 4. The following are all functions that are single transformations of $y = \arcsin x$. Write down the equation in each case.

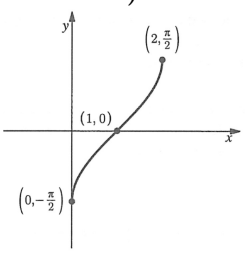
a)



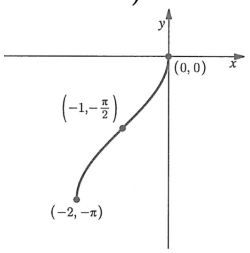
b)



c)



d)



2.14 — Question 5. Sketch the graph of $y = -\arcsin(x+1)$ in radians, identifying the coordinates of any endpoints.

2.14 — Question 6. Sketch the graph of $y = \frac{\pi}{2} - \arcsin x$ in radians, identifying the coordinates of any endpoints.

2.14 — Question 7. It is given that $f(x) = \arcsin x$, where $-1 \leq x \leq 1$. Find the exact value for which $\pi - 3f(2x - 3) = 0$.

2.14 — Question 8. Prerequisite knowledge for Q8: Domain and Range and Inverse Functions It is given that $f(x) = 2 - 3\sin x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.

a) Show that $f^{-1}(x) = \arcsin\left(\frac{2-x}{3}\right)$.

b) State the domain and range of $y = f^{-1}(x)$.

2.14 — Question 9. Sketch the following graphs, in radians, identifying the coordinates of any endpoints.

a) $y = \frac{5}{2}\arccos x$

b) $y = \frac{1}{5}\arccos x$

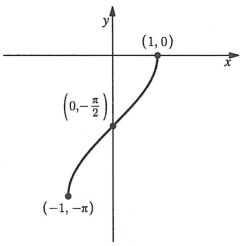
2.14 — Question 10. Sketch the following graphs, in radians, identifying the coordinates of any endpoints.

a) $y = \arccos 4x$

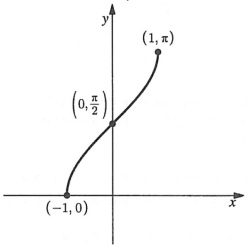
b) $y = \arccos \frac{4}{5}x$

2.14 — Question 11. The following are all functions that are single transformations of $y = \arccos x$. Write down the equation in each case.

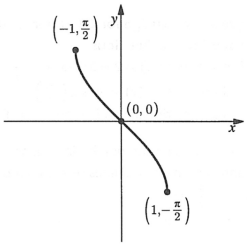
a)



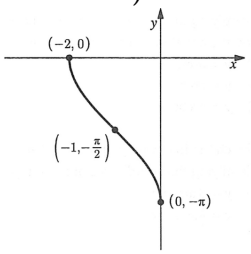
b)



c)



d)



2.14 — Question 12. Sketch the graph of $y = -\arccos(x-1)$ in radians, identifying the coordinates of any endpoints.

2.14 — Question 13. Sketch the graph of $y = \arccos 2x - \pi$ in radians, identifying the coordinates of any endpoints.

2.14 — Question 14. It is given that $g(x) = \arccos x$, where $-1 \leq x \leq 1$. Find the exact value for which $6\pi - 9g(0.2x + 3) = 0$.

2.14 — Question 15.

a) Solve the equation $\arccos x = \frac{\pi}{3}$.

b) Hence find the exact value of $\arcsin x$.

2.14 — Question 16. Sketch the following graphs, in radians, identifying the equations of any asymptotes:

a) $y = 8 \arctan x$

b) $y = \frac{1}{6} \arctan x$

2.14 — Question 17. Sketch the following graphs, in radians, identifying the equations of any asymptotes:

a) $y = 2 + 3 \arctan x$

b) $y = 5 - 2 \arctan x$

2.14 — Question 18. Sketch the following graphs, in radians, identifying any endpoints and the equations of any asymptotes:

a) $y = 3\arcsin x + \pi$

b) $y = \arccos(\frac{1}{2}x - 2)$

c) $y = 2\arctan x - \frac{\pi}{3}$

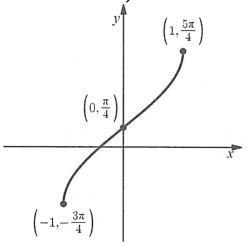
d) $y = \arcsin(4x + 5)$

e) $y = \frac{2\pi}{3} - \arccos x$

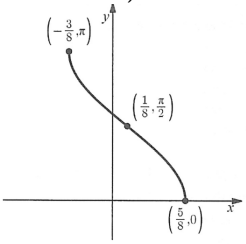
f) $y = 3\pi - \arctan x$

2.14 — Question 19. Write down **two** suitable equations for the following graphs:

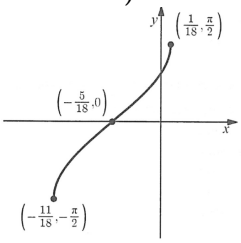
a)



b)



c)



2.14 — Question 20. Solve the equation $\arcsin x = \arccos x$.

2.14 — Question 21. Prerequisite knowledge for Q21: Double Angle Formulae

a) Solve the equation $\arcsin x = 2\arccos x$.

b) Solve the equation $2\arcsin x = \arccos x$.

2.14 — Question 22. Given $\arcsin x = \frac{\pi}{5}$, show that $x = \sqrt{\frac{5-\sqrt{5}}{8}}$.

The remaining questions in this exercise are all in radians and are to be completed without a calculator.

2.14 — Question 23. Find

a) $\arcsin 1$

b) $\arccos(-1)$

c) $\arcsin 0$

d) $\arccos 1$

2.14 — Question 24. Find

a) $\arcsin \frac{\sqrt{2}}{2}$

b) $\arccos \frac{\sqrt{2}}{2}$

c) $\arcsin\left(-\frac{\sqrt{2}}{2}\right)$

d) $\arccos\left(-\frac{\sqrt{2}}{2}\right)$

2.14 — Question 25. Find

a) $\arctan 1$

b) $\arctan(-1)$

c) $\arctan \sqrt{3}$

d) $\arctan\left(-\frac{\sqrt{3}}{3}\right)$

2.14 — Question 26. Find

a) $\arcsin\left(\sin\frac{\pi}{2}\right)$

b) $\arccos\left(\cos\left(-\frac{\pi}{2}\right)\right)$

c) $\arcsin\left(\sin\frac{3\pi}{2}\right)$

d) $\arccos\left(\cos\left(-\frac{3\pi}{2}\right)\right)$

e) $\arcsin\left(\sin\frac{\pi}{3}\right)$

f) $\arccos\left(\cos\frac{2\pi}{3}\right)$

g) $\arcsin\left(\sin\frac{5\pi}{3}\right)$

h) $\arccos\left(\cos\left(-\frac{\pi}{3}\right)\right)$

i) $\arctan\left(\tan\frac{\pi}{4}\right)$

j) $\arctan\left(\tan\frac{2\pi}{3}\right)$

2.14 — Question 27. Find

a) $\sin\left(\arcsin\frac{\sqrt{2}}{2}\right)$

b) $\cos\left(\arccos\left(-\frac{\sqrt{2}}{2}\right)\right)$

c) $\sin\left(\arccos\frac{1}{2}\right)$

d) $\cos\left(\arcsin\frac{1}{2}\right)$

e) $\sin\left(\arccos\frac{\sqrt{3}}{2}\right)$

f) $\cos\left(\arcsin\left(-\frac{\sqrt{3}}{2}\right)\right)$

g) $\tan\left(\arcsin\left(-\frac{1}{2}\right)\right)$

2.15 — Question 1. Sketch the following graphs for $0° < x < 360°$:

a) $y = \operatorname{cosec} x$

b) $y = \sec x$

c) $y = \cot x$

2.15 — Question 2. Sketch the following graphs for $0 < x < 2\pi$:

a) $y = \operatorname{cosec} x$

b) $y = \sec x$

c) $y = \cot x$

2.15 — Question 3. Sketch the following graphs for $0° < x < 360°$:

a) $y = 2\operatorname{cosec} x$
b) $y = \frac{1}{3}\sec x$
c) $y = -\cot x$
d) $y = \operatorname{cosec}(x - 90°)$
e) $y = \sec(x + 60°)$
f) $y = \cot(x - 270°)$

2.15 — Question 4. Sketch the following graphs for $0 < x < 2\pi$:

a) $y = \frac{1}{4}\operatorname{cosec} x$
b) $y = \frac{4}{3}\sec x$
c) $y = \cot(-x)$
d) $y = \operatorname{cosec}(x + \frac{\pi}{2})$
e) $y = \sec(x - \frac{\pi}{3})$
f) $y = \cot(x + \frac{2\pi}{3})$

2.15 — Question 5. Prerequisite knowledge for Q5–Q6: Combinations of Graph Transformations Sketch the following graphs for $0° < x < 360°$:

a) $y = 1 + 3\operatorname{cosec} x$
b) $y = 2 - \sec x$
c) $y = -\cot(x - 30°)$
d) $y = \operatorname{cosec}(2x - 90°)$
e) $y = \sec(2x + 60°)$
f) $y = \cot(\frac{1}{2}x - 30°)$

2.15 — Question 6. Sketch the following graphs for $0 < x < 2\pi$:

a) $y = \frac{1}{2}\operatorname{cosec} x - 1$
b) $y = \frac{1}{3}\sec x - 2$
c) $y = \cot(-2x)$
d) $y = \operatorname{cosec}(\frac{1}{2}x + \frac{\pi}{4})$
e) $y = \sec(\frac{1}{2}x - \frac{\pi}{3})$
f) $y = \cot(2x + \frac{\pi}{3})$

2.15 — Question 7. Solve the following equations:

a) $\operatorname{cosec}\theta = 2,\ 0° < \theta < 360°$

b) $\sec\theta = 3,\ 0 < \theta < 2\pi$

c) $\cot\theta = 4,\ 0° < \theta < 360°$

d) $\operatorname{cosec}\theta = -\frac{5}{4},\ 0 < \theta < 2\pi$

e) $\sec\theta = -\frac{9}{2},\ 0° < \theta < 360°$

f) $\cot\theta = -\frac{18}{5},\ 0 < \theta < 2\pi$

2.15 — Question 8. State the values of x for which $y = \operatorname{cosec} x$ is an increasing function, for $0 < x < 2\pi$

2.15 — Question 9. State the values of x for which $y = \sec x$ is a decreasing function, for $0° < x < 360°$

2.15 — Question 10. It is given that $\operatorname{cosec} x = \frac{7}{2}$ and x is acute. Find the exact values of:

a) $\sin x$
b) $\cos x$
c) $\tan x$
d) $\sec x$
e) $\cot x$

2.15 — Question 11. It is given that $\sec x = \frac{9}{4}$ and x is acute. Find the exact values of:

a) $\cos x$
b) $\sin x$
c) $\tan x$
d) $\operatorname{cosec} x$
e) $\cot x$

2.15 — Question 12. It is given that $\cot x = \frac{12}{11}$ and x is acute. Find the exact values of:

a) $\tan x$
b) $\sin x$
c) $\cos x$
d) $\operatorname{cosec} x$
e) $\sec x$

2.15 — Question 13. It is given that $\operatorname{cosec} x = \frac{10}{9}$ and x is obtuse. Find the exact values of:

a) $\sin x$
b) $\cos x$
c) $\tan x$
d) $\sec x$
e) $\cot x$

2.15 — Question 14. It is given that $\sec x = -\frac{12}{5}$ and $180° < x < 270°$. Find the exact values of:

a) $\cos x$
b) $\sin x$
c) $\tan x$
d) $\operatorname{cosec} x$
e) $\cot x$

2.15 — Question 15. It is given that $\cot x = -\frac{2}{7}$ and $270° < x < 360°$. Find the exact values of:

a) $\tan x$
b) $\sin x$
c) $\cos x$
d) $\operatorname{cosec} x$
e) $\sec x$

2.15 — Question 16. It is given that $8\sec^2 x - 14\sec x + 3 = 0$ and x is acute. Find the exact values of:

a) $\sec x$
b) $\cos x$
c) $\sin x$
d) $\tan x$
e) $\operatorname{cosec} x$
f) $\cot x$

2.15 — Question 17. Without using any calculus, state the maximum value of the curve with equation $y = \frac{2}{\operatorname{cosec}^2 x + 2}$

2.15 — Question 18. Write the following trigonometric expressions in terms of $\tan x$, $\operatorname{cosec} x$, $\sec x$ and $\cot x$ only:

a) $\frac{\sin x}{\cos^2 x}$

b) $\frac{\cos x}{\sin^2 x}$

2.15 — Question 19. Simplify the following trigonometric expressions as far as possible:

a) $\operatorname{cosec} x \sin x$

b) $\sec x \cos x$

c) $\cot x \tan x$

2.15 — Question 20. Simplify the following trigonometric expressions as far as possible, writing your answers in terms of $\sin x$ and $\cos x$ only:

a) $\operatorname{cosec} x \sec x$

b) $\operatorname{cosec} x \cot x$

c) $\sec x \cot x$

d) $\operatorname{cosec} x \sec x \cot x$

2.15 — Question 21. Simplify the following trigonometric expressions as far as possible:

a) $\frac{\operatorname{cosec} x}{\sec x}$
b) $\frac{\operatorname{cosec} x}{\cot x}$
c) $\frac{\sec x}{\operatorname{cosec} x}$
d) $\frac{\sec x}{\cot x}$
e) $\frac{\cot^2 x}{\sec x}$
f) $\frac{\cot^2 x \sin^3 x}{\operatorname{cosec} x \sec x}$

2.15 — Question 22. Solve the following equations:

a) $\operatorname{cosec}\theta = \sec\theta,\ 0° < \theta < 360°$
b) $\cot\theta = -\sec\theta,\ 0 < \theta < 2\pi$
c) $\sec\theta = \cot\theta,\ 0° < \theta < 360°$
d) $\cot\theta = \tan\theta,\ 0 < \theta < 2\pi$

2.15 — Question 23.

a) Sketch the graphs of $y = \operatorname{cosec} x$ and $y = 2\sec x$ on the same axes in radians between 0 and 2π

b) Determine the coordinates of where the two graphs intersect.

2.15 — Question 24. State the values of k for which $\sec x - 3 + k = 0$ has real solutions.

2.15 — Question 25. Starting from $\sin^2 x + \cos^2 x \equiv 1$

a) derive an identity for $\cot x$ and $\operatorname{cosec} x$,

b) derive an identity for $\tan x$ and $\sec x$

2.15 — Question 26. Simplify the following trigonometric expressions:

a) $\frac{1+\cot^2 x}{\operatorname{cosec}^2 x}$

b) $\frac{1+\tan^2 x}{\sec^2 x}$

c) $\frac{1+\tan^2 x}{1+\cot^2 x}$

d) $\frac{2+2\cot^2 x}{5+5\tan^2 x}$

2.15 — Question 27. Solve the following equations:

a) $\cot^2\theta + 3\operatorname{cosec}\theta - 3 = 0,\ 0° < \theta < 360°$

b) $\tan^2\theta - \sec\theta - 11 = 0,\ 0 < \theta < 2\pi$

c) $\operatorname{cosec}^2\theta + 5\cot\theta - 15 = 0,\ 0° < \theta < 360°$

d) $\sec^2\theta + 6\tan\theta - 28 = 0,\ 0 < \theta < 2\pi$

2.15 — Question 28. Solve the following equations:

a) $3\operatorname{cosec}^2\theta + 2\cot\theta - 4 = 0,\ 0° < \theta < 360°$

b) $4\sec^2\theta - 4\tan\theta - 19 = 0,\ 0 < \theta < 2\pi$

c) $6\cot^2\theta + 23\operatorname{cosec}\theta + 13 = 0,\ 0° < \theta < 360°$

d) $18\tan^2\theta + 9\sec\theta - 2 = 0,\ 0 < \theta < 2\pi$

2.15 — Question 29. Show that $3\tan^2 x + 4k\sec x + \frac{4k^2}{3} + 3$ can be written in the form $3\left(\sec x + \frac{2k}{3}\right)^2$

2.15 — Question 30. It is given that $10\tan^2 x + 17\sec x + 16 = 0$ and x is obtuse. Find the exact values of:

a) $\sec x$
b) $\cos x$
c) $\sin x$
d) $\tan x$
e) $\operatorname{cosec} x$
f) $\cot x$

2.15 — Question 31. It is given that $16\cot^2 x + 32\operatorname{cosec} x + 31 = 0$ and $\frac{3\pi}{2} < x < 2\pi$. Find the exact values of:

a) $\operatorname{cosec} x$
b) $\sin x$
c) $\cos x$
d) $\tan x$
e) $\sec x$
f) $\cot x$

2.15 — Question 32. Prove there are no real solutions to the equation $12\cot^2\theta - 5\operatorname{cosec}\theta + 10 = 0$

2.15 — Question 33. It is given that $\mathrm{p}(x) = 12x^3 + 8x^2 - 3x - 2$

a) Show that $(3x + 2)$ is a factor of $\mathrm{p}(x)$

b) Write $\mathrm{p}(x)$ as a product of three linear factors.

c) Hence prove that there are no real solutions to the equation $12\,\mathrm{cosec}^3\,\theta + 8\cot^2\theta - 3\,\mathrm{cosec}\,\theta + 6 = 0$

2.15 — Question 34. Prove the following trigonometric identities:

a) $\cot x + \cos x \equiv \cot x(1 + \sin x)$

b) $\cot x + \sec x \equiv \mathrm{cosec}\,x \sec x(\sin x + \cos^2 x)$

c) $\cot x + \sin x \equiv \mathrm{cosec}\,x(\sin^2 x + \cos x)$

d) $\cot x + \tan x \equiv \mathrm{cosec}\,x \sec x$

2.15 — Question 35. Prove the following trigonometric identities:

a) $\sec x - \cos x \equiv \sin x \tan x$

b) $\operatorname{cosec} x - \sin x \equiv \cos x \cot x$

c) $\tan x - \cot x \equiv \operatorname{cosec} x \sec x(2\sin^2 x - 1)$

d) $\tan x - \operatorname{cosec} x \equiv \operatorname{cosec} x(\sin x \tan x - 1)$

2.15 — Question 36. Prove the following trigonometric identities:

a) $\sin^2 x - \cot^2 x \equiv \operatorname{cosec}^2 x(\sin^4 x - \cos^2 x)$

b) $\cos^2 x - \cot^2 x \equiv -\cos^2 x \cot^2 x$

c) $\sec^2 x - \operatorname{cosec}^2 x \equiv \operatorname{cosec}^2 x \sec^2 x(2\sin^2 x - 1)$

d) $\tan^2 x - \cot^2 x \equiv \operatorname{cosec}^2 x \sec^2 x(2\sin^2 x - 1)$

2.15 — Question 37. Prove the following trigonometric identities:

a) $(\sec x + \cot x)^2 = \tan^2 x + 2\operatorname{cosec} x + \operatorname{cosec}^2 x$

b) $(\sec x + \operatorname{cosec} x)^2 \equiv \tan^2 x + \cot^2 x + 2(\tan x + \cot x + 1)$

c) $(\operatorname{cosec} x + \cot x)^2 \equiv 2\cot^2 x + 1 + 2\cos x(\cot^2 x + 1)$

d) $(\tan x + \sec x)^2 \equiv 2\sec^2 x(\sin x + 1) - 1$

e) $(\tan x + \sec x)^2 \equiv \frac{1+\sin x}{1-\sin x}$

2.15 — Question 38.

a) Prove the following trigonometric identity:

$$\frac{\sin\theta}{1-\cos\theta} - \operatorname{cosec}\theta \equiv \cot\theta$$

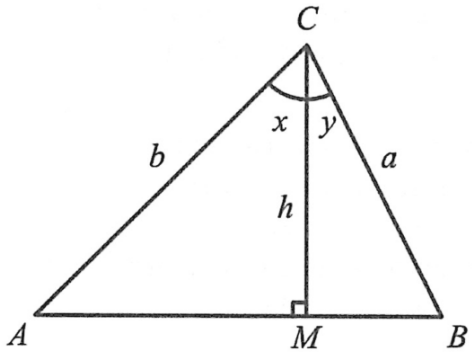
b) Hence solve the equation

$$\frac{\sin\theta}{1-\cos\theta} = 10 + \cot\theta$$

for $0 < \theta < 2\pi$

2.15 — Question 39. Prerequisite knowledge for Q39: Double Angle Formulae Solve the equation $\operatorname{cosec}\theta = 10\cos\theta$ for $0 < \theta < 2\pi$

2.16 — Question 1. The diagram below shows the triangle ABC. It is given that angle ACM is x degrees and angle BCM is y degrees. The length of BC is a and the length of AC is b.



a) Write h in terms of b and x.

b) Write h in terms of a and y.

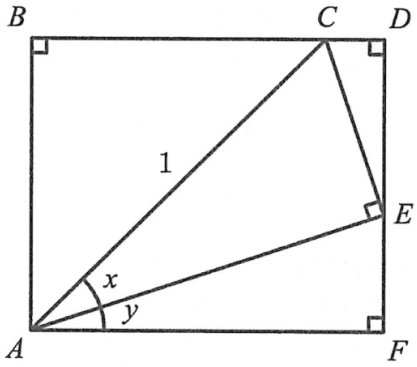
c) Find the area of triangles ACM and BCM.

d) Find the area of triangle ABC.

e) Hence show that

$$\sin(x+y) = \sin x \cos y + \cos x \sin y.$$

2.16 — Question 2. The diagram below shows the rectangle $ABDF$. It is given that angle CAE is x degrees and angle EAF is y degrees. It is also given that the length AC is 1 unit.



a) Find the lengths of AE & CE in terms of x.

b) Find the angles ACE, AEF, CED, ECD, BCA and BAC in terms of x and y.

c) Find the lengths of BC and AB in terms of x and y.

d) Find the lengths of AF and EF in terms of x and y.

e) Find the lengths of DE and CD in terms of x and y.

f) Use the fact that $AB = DE + EF$ and $BC = AF - CD$ to write down two compound angle formulae.

2.16 — Question 3. Given $\sin(x + y) = \sin x \cos y + \cos x \sin y$, show that $\sin(x - y) = \sin x \cos y - \cos x \sin y$.

2.16 — Question 4. Given $\cos(x+y) = \cos x \cos y - \sin x \sin y$, show that $\cos(x-y) = \cos x \cos y + \sin x \sin y$.

2.16 — Question 5. By writing $\tan(x+y)$ as $\frac{\sin(x+y)}{\cos(x+y)}$, show that

$$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}.$$

2.16 — Question 6. Given $\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$, show that

$$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}.$$

2.16 — Question 7. Write the following in the form $\sin k$, where k is an acute angle:

a) $\sin 30° \cos 45° + \cos 30° \sin 45°$

b) $\sin 35° \cos 20° - \cos 35° \sin 20°$

c) $\sin 70° \cos 50° - \cos 70° \sin 50°$

d) $\sin 70° \cos 50° + \cos 70° \sin 50°$

2.16 — Question 8. Write the following in the form $\cos k$, where k is an acute angle:

a) $\cos 50° \cos 30° + \sin 50° \sin 30°$

b) $\cos 68° \cos 13° + \sin 68° \sin 13°$

c) $\cos 10° \cos 54° - \sin 10° \sin 54°$

d) $\cos 300° \cos 50° - \sin 300° \sin 50°$

2.16 — Question 9.

a) i) Write down the exact values of $\sin 30°, \sin 45°$ and $\sin 60°$

ii) Write down the exact values of $\cos 30°, \cos 45°$ and $\cos 60°$

b) Hence use a compound angle formula to find the exact value of:

i) $\sin 75°$

ii) $\sin 15°$

iii) $\cos 105°$

iv) $\cos 15°$

2.16 — Question 10.

a) Write down the exact values of $\tan 30°$, $\tan 45°$ and $\tan 60°$.

b) Hence use a compound angle formula to find the exact value of:

i) $\tan 75°$

ii) $\tan 15°$

2.16 — Question 11. Use the compound angle formulae to simplify each of these:

a) $\sin(x + 90°)$

b) $\cos(x - 90°)$

c) $\cos(x - 180°)$

d) $\sin(x - 270°)$

2.16 — Question 12. Write the following in the form $\sin k$, where k is an acute angle:

a) $\cos 40° \cos 32° - \sin 40° \sin 32°$

b) $\cos 128° \cos 125° + \sin 128° \sin 125°$

2.16 — Question 13. Write the following in the form $\cos k$, where k is an acute angle:

a) $\sin 96° \cos 5° - \cos 96° \sin 5°$

b) $\sin 25° \cos 42° + \cos 25° \sin 42°$

2.16 — Question 14. It is given that $\cos x = \frac{2}{9}$ and $\cos y = \frac{6}{7}$ and that x and y are acute.
Find the exact value of:

a) $\cos(x - y)$

b) $\sin(x - y)$

2.16 — Question 15. It is given that $\tan 85° = b$. Show that

$$\tan 130° = \frac{1 + b}{1 - b}.$$

2.16 — Question 16. It is given that $\cos 55° = m$. Show that

$$\cos 5° = \frac{m + \sqrt{3 - 3m^2}}{2}.$$

2.16 — Question 17. Find the minimum value of the following expressions and the positive value of θ, in degrees, for which the minimum first occurs:

a) $\sin\theta \cos 60° + \cos\theta \sin 60°$

b) $2\cos\theta \cos 35° + 2\sin\theta \sin 35°$

c) $4\sin\theta \cos 42° - 4\cos\theta \sin 42°$

2.16 — Question 18. Prerequisite knowledge for Q18–Q19: Radians Find the minimum value of the following expressions and the positive value of θ, in radians, for which the minimum first occurs:

a) $\sin\theta\cos\frac{\pi}{4}+\cos\theta\sin\frac{\pi}{4}$

b) $3\cos\theta\cos\frac{5\pi}{4}+3\sin\theta\sin\frac{5\pi}{4}$

c) $-2\cos\theta\sin 1.6+2\sin\theta\cos 1.6$

2.16 — Question 19. Find the maximum value of the following expressions and the positive value of θ, in radians, for which the maximum first occurs:

a) $\sin 3\theta\cos 2\theta+\cos 3\theta\sin 2\theta$

b) $9\cos 5\theta\cos 2\theta+9\sin 5\theta\sin 2\theta$

c) $-\frac{2}{3}\cos 8\theta\sin 2\theta+\frac{2}{3}\sin 8\theta\cos 2\theta$

2.16 — Question 20. Solve the following equations:

a) $\sin\theta\cos 20° + \cos\theta\sin 20° = 0.5$ for $0° \leq \theta < 360°$

b) $\cos\theta\cos 40° + \sin\theta\sin 40° = 0.34$ for $0° \leq \theta < 360°$

2.16 — Question 21. Write the following in the form $\tan k$, where k is an acute angle:

a) $\frac{\tan 200° + \tan 34°}{1 - \tan 200° \tan 34°}$

b) $\frac{\tan 18° - \tan 130°}{1 + \tan 18° \tan 130°}$

2.16 — Question 22. Solve the following equations:

a) $\frac{\tan\theta+\tan 20^\circ}{1-\tan\theta\tan 20^\circ} = 4$ for $0^\circ \leq \theta < 360^\circ$

b) $\frac{\tan 3\theta-\tan 33^\circ}{1+\tan 3\theta\tan 33^\circ} = -6$ for $0^\circ \leq \theta < 120^\circ$

2.16 — Question 23. Simplify the following expressions:

a) $\sin(x+45^\circ)+\cos(x+45^\circ)$

b) $\sin(x+135^\circ)-\cos(x+135^\circ)$

c) $\sin(x-75^\circ)-\cos(x-75^\circ)$

2.16 — Question 24. Prerequisite knowledge for Q24: Radians Simplify the following expressions:

a) $\sin(x - \frac{\pi}{3}) + \cos(\frac{\pi}{3} - x)$

b) $\sin(\frac{\pi}{3} - x) - \cos(\frac{4\pi}{3} - x)$

c) $-\sin(\frac{\pi}{4} - x) + \cos(\frac{2\pi}{3} - x)$

2.16 — Question 25. Solve the following equations:

a) $\sin(\theta + 30°) = \cos\theta$ for $0° < \theta < 360°$

b) $\sin(\theta + 30°) = \sin\theta$ for $0° < \theta < 360°$

c) $\cos(\theta + 30°) = \sin\theta$ for $0° < \theta < 360°$

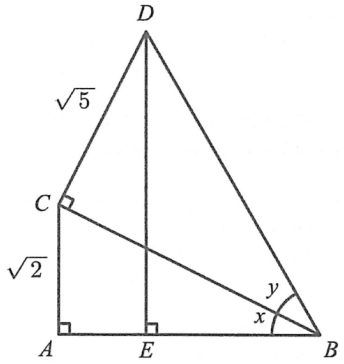
2.16 — Question 26. Solve the following equations:

a) $\sin(\theta - 30°) = \cos(\theta + 45°)$ for $0° < \theta < 360°$

b) $\cos(\theta - 60°) = \cos(\theta - 45°)$ for $0° < \theta < 360°$

c) $\cos(\theta - 120°) = \sin(\theta + 60°)$ for $0° < \theta < 360°$

2.16 — Question 27. In the diagram below, angle ABC is x degrees and angle CBD is y degrees. The lengths of AC and CD are $\sqrt{2}$ and $\sqrt{5}$ respectively. Show that $DE = \sqrt{2} + \sqrt{5}\cos x$.



2.16 — Question 28. Prove the following trigonometric identity

$$\frac{\cos(x+y)}{\cos x \cos y} \equiv 1 - \tan x \tan y.$$

2.16 — Question 29. Solve the equation

$$\frac{\tan\theta - \tan 30^\circ}{1 + \tan\theta \tan 30^\circ} = 2$$

for $0^\circ < \theta < 360^\circ$.

2.16 — Question 30.

a) Solve the equation

$$3\cos(x - 45°) = 2\sin(x + 60°)$$

for $0° < x < 360°$.

b) Hence solve the equation

$$3\cos(3\theta - 45°) = 2\sin(3\theta + 60°)$$

for $0° < \theta < 120°$.

2.16 — Question 31. It is given that $\sin x = \frac{2}{\sqrt{14}}$ and $\cos y = \frac{3}{7}$ and that x is obtuse and y is acute. Show that the exact value of $\tan(x - y)$ is $\frac{13\sqrt{10}}{5}$.

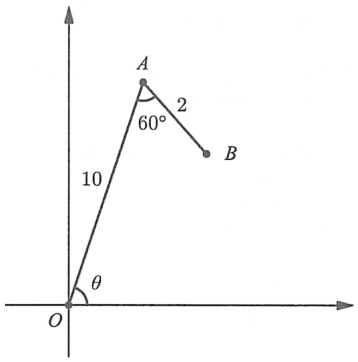
2.16 — Question 32. It is given that $\tan x = \frac{1}{2}$ and that x is an acute angle. Given that $\tan(x+y) = 13$, find the exact value of $\tan y$.

2.16 — Question 33. Prove the following trigonometric identity

$$2\sin(x+45^\circ)\sin(x-45^\circ) \equiv \sin^2 x - \cos^2 x.$$

2.16 — Question 34. The diagram below shows the rod OAB which is pivoted at O. The length of OA is 10 units and the length of AB is 2 units. Taking O as the point with coordinates $(0, 0)$, show that the coordinates of B are

$$(9\cos\theta + \sqrt{3}\sin\theta,\ 9\sin\theta - \sqrt{3}\cos\theta).$$



2.16 — Question 35. Prerequisite knowledge for Q35–Q36: Small Angle Approximation By considering compound angle formulae, show that

$$\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right).$$

Hence, prove by first principles that

$$\frac{\mathrm{d}}{\mathrm{d}x}(\cos x) = -\sin x.$$

2.16 — Question 36. By considering compound angle formulae, show that

$$\sin A - \sin B = 2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right).$$

Hence, prove by first principles that

$$\frac{\mathrm{d}}{\mathrm{d}x}(\sin x) = \cos x.$$

2.17 — Question 1. Use the Compound Angle Formulae $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ to find the Double Angle Formula for $\sin 2x$.

2.17 — Question 2. Use the Compound Angle Formulae $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ to find a simplified form for $\cos 2x$.

2.17 — Question 3. Write down $\cos 2x$ just in terms of $\cos x$.

2.17 — Question 4. Write down $\cos 2x$ just in terms of $\sin x$.

2.17 — Question 5. Use the Compound Angle Formulae $\tan(A \pm B) = \dfrac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$ to find a simplified form for $\tan 2x$.

2.17 — Question 6. Use $\tan 2x = \frac{\sin 2x}{\cos 2x}$ to find the same result found in question 5.

2.17 — Question 7. Simplify the following expressions:

a) $2\sin 2x \cos 2x$

b) $2\cos^2 3x - 1$

c) $1 - 2\sin^2 4x$

d) $4\sin \frac{1}{2}x \cos \frac{1}{2}x$

e) $6\cos^2 \frac{1}{3}x - 6\sin^2 \frac{1}{3}x$

f) $6\cos^2 \frac{3}{2}x - 3$

2.17 — Question 8. Solve the following equations:

a) $\sin 2\theta - 2\sin\theta = 0$ for $0 \leq \theta < 360°$

b) $\sin 2\theta + 5\sin\theta = 0$ for $0 \leq \theta < 2\pi$

c) $3\sin 2\theta = 2\cos\theta$ for $0 \leq \theta < 360°$

d) $8\sin 2\theta + 5\cos\theta = 0$ for $0 \leq \theta < 2\pi$

2.17 — Question 9. Solve the following equations:

a) $\cos 2\theta + \cos\theta = 0$ for $0 \leq \theta < 360°$

b) $3\cos 2\theta = 7\cos\theta$ for $0 \leq \theta < 2\pi$

c) $2\cos 2\theta + 2\sin\theta = 0$ for $0 \leq \theta < 360°$

d) $2\cos 2\theta = 9\sin\theta$ for $0 \leq \theta < 2\pi$

2.17 — Question 10. Prerequisite knowledge for Q10: Reciprocal Trigonometric Functions Solve the following equations:

a) $2\sin 2\theta \operatorname{cosec}\theta = 1$ for $0 < \theta < 360°$

b) $5\sin 2\theta \sec\theta = 2$ for $0 < \theta < 2\pi$

c) $\cos 2\theta \operatorname{cosec}\theta = 3$ for $0 < \theta < 360°$

d) $9\cos 2\theta \sec\theta = 4$ for $0 < \theta < 2\pi$

2.17 — Question 11. It is given that $\sin x = \frac{2}{5}$ and that x is acute.
Find the exact value of $\sin 2x$.

2.17 — Question 12. It is given that $\sin x = \frac{3}{8}$ and that x is acute.
Find the exact value of $\cos 2x$.

2.17 — Question 13. It is given that $\cos x = -\frac{2}{9}$ and that x is obtuse.
Find the exact value of $\tan 2x$.

2.17 — Question 14. Given that $10\sin^2 x + 5\sin 2x = 4$, show that $3\tan^2 x + 5\tan x - 2 = 0$.

2.17 — Question 15. Solve the following equations:

a) $34\sin^2\theta - \sin 2\theta = 30$ for $0° \leq \theta < 360°$

b) $22\cos^2\theta - 13\sin 2\theta = 10$ for $0 < \theta < 2\pi$

2.17 — Question 16. Prerequisite knowledge for Q16–Q21: Reciprocal Trigonometric Functions Prove the following trigonometric identities:

a) $\tan 2x + \cot x \equiv \cot x \sec 2x$

b) $\tan 2x + \sin 2x \equiv \tan 2x(\cos 2x + 1)$

c) $\tan 2x + \cot 2x \equiv 2\operatorname{cosec} 4x$

2.17 — Question 17. Prove the following trigonometric identities:

a) $\cot 2x - \cos 2x \equiv \cot 2x(1 - \sin 2x)$

b) $\cot 2x + \operatorname{cosec} 2x \equiv \operatorname{cosec} 2x(1 + \cos 2x)$

c) $\cot 2x + \operatorname{cosec} 2x \equiv \cot x$

2.17 — Question 18. Prove the following trigonometric identity:

$$\tan 2x + \operatorname{cosec} 2x \equiv 2 \operatorname{cosec} 4x(\sin^2 2x + \cos 2x)$$

2.17 — Question 19. Prove the following trigonometric identities:

a) $\frac{2 \tan 2x}{1+\tan^2 2x} \equiv \sin 4x$

b) $(\sin 2x + \cos 2x)^2 \equiv 1 + \sin 4x$

c) $\frac{2 \sin x}{\cos 2x+1} - \frac{2 \sin x}{\cos 2x-1} \equiv \operatorname{cosec} x \sec^2 x$

2.17 — Question 20. Prove the following trigonometric identities:

a) $\frac{1-\sin 2x}{1+\sin 2x} \equiv \frac{2}{(\sin x+\cos x)^2} - 1$

b) $\frac{1-\cos 2x}{1+\sin 2x} \equiv \frac{2\sin^2 x}{(\sin x+\cos x)^2}$

c) $\frac{1+\cos 2x-\sin 2x}{1+\sin 2x-\cos 2x} \equiv \frac{\cos x(\cot x-1)}{\sin x+\cos x}$

2.17 — Question 21.

a) Prove the following trigonometric identity:

$$\operatorname{cosec}^2 x \equiv \frac{\cos 2x + 3\sin^2 x + \cos^2 x}{2\sin^2 x}.$$

b) Hence solve

$$(\cot^2\theta - 1)\operatorname{cosec}^2\theta = \frac{\cos 2\theta + 3\sin^2\theta + \cos^2\theta}{6\sin^2\theta}$$

for $0 < \theta < 2\pi$.

2.17 — Question 22.

a) Solve

$$11\sin 2x = 2\cos x + 6\cos^3 x$$

for $0 < x < 2\pi$.

b) Solve

$$11\sin(6\theta - \tfrac{\pi}{2}) = 2\cos(3\theta - \tfrac{\pi}{4}) + 6\cos^3(3\theta - \tfrac{\pi}{4})$$

for $0 < \theta < \frac{2\pi}{3}$.

2.17 — Question 23. Prove the following trigonometric identities:

a) $\sqrt{2}\sin(2x + \frac{\pi}{4}) \equiv \cos^2 x + 2\sin x\cos x - \sin^2 x$

b) $2\sqrt{3}\sin(\frac{\pi}{3} - 2x) \equiv 3\cos^2 x - 2\sqrt{3}\sin x\cos x - 3\sin^2 x$

c) $\sin(\frac{\pi}{4} - 2x) + \cos(2x + \frac{\pi}{4}) + 2\sqrt{2}\sin^2 x \equiv \sqrt{2}(\cos x - \sin x)^2$

d) $\frac{1}{2}\cos(\frac{\pi}{6} - 2x) - \frac{1}{2}\sin(\frac{\pi}{3} - 2x) \equiv \sin x\cos x$

2.17 — Question 24. Prove the following trigonometric identity:

$$(\sin(x + \tfrac{\pi}{4}) + \cos(x + \tfrac{\pi}{4}))^2 \equiv \cos 2x + 1$$

2.17 — Question 25. Prove the following trigonometric identity:

$$(\sin(2x + \tfrac{\pi}{4}) - \cos(2x + \tfrac{\pi}{4}))^2 \equiv 1 - \cos 4x$$

2.17 — Question 26. Use the Compound Angle Formulae $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ to show that $\sin 3x \equiv 3\sin x - 4\sin^3 x$.

2.17 — Question 27. Use the Compound Angle Formulae $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ to show that $\cos 3x \equiv 4\cos^3 x - 3\cos x$.

2.17 — Question 28. Prove the following trigonometric identity:

$$\sqrt{2}\sin(\tfrac{\pi}{4} - x)(2\sin 2x + 1) \equiv \sin 3x + \cos 3x$$

2.17 — Question 29. Find the period of $\mathrm{f}(x) = 5 - \cos^2 x$, $x \in \mathbb{R}$.

2.17 — Question 30. Prove the following trigonometric identity:

$$\frac{\sin 3x}{\sin x} + \frac{\cos 3x}{\cos x} \equiv 4\cos 2x$$

2.17 — Question 31. Use the Compound Angle Formulae $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$ to find $\tan 3x$ in terms of $\tan x$ only.

2.17 — Question 32. Solve the equation $\tan 3\theta + \tan \theta = 0$ for $0° < \theta < 360°$.

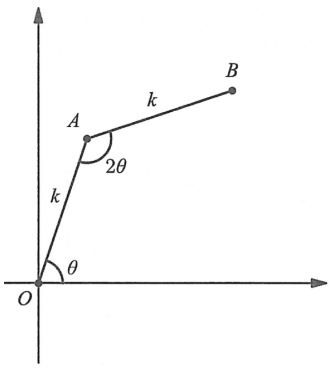
2.17 — Question 33. Solve $\sin \frac{2\theta}{3} \cos \frac{2\theta}{3} = \frac{1}{2}$ for $0 < \theta < 2\pi$.

2.17 — Question 34. Solve $\cos^2 \frac{3\theta}{2} - \sin^2 \frac{3\theta}{2} = \frac{1}{3}$ for $0° < \theta < 360°$.

2.17 — Question 35. The diagram below shows the rod OAB which is pivoted at O. The lengths of OA and AB are both k units. Taking O as the point with coordinates $(0, 0)$,

a) show that the coordinates of B are $(4k \sin^2 \theta \cos \theta,\ 2k \sin \theta - 4k \sin \theta \cos^2 \theta)$

b) Find the distance of B from O when $\theta = 60°$ in terms of k.



2.17 — Question 36. Prove the following trigonometric identity:

$$\sin 4x \equiv 4 \sin x \cos^3 x - 4 \sin^3 x \cos x$$

2.17 — Question 37. Prove the following trigonometric identity:

$$\cos 4x \equiv 8 \cos^4 x - 8 \cos^2 x + 1$$

2.17 — Question 38. Prerequisite knowledge for Q38: Reciprocal Trigonometric Functions Prove the following trigonometric identity:

$$\frac{\sin 4x}{\cos x} + \frac{\cos 4x}{\sin x} \equiv (2\cos 2x - 1)\operatorname{cosec} x$$

2.18 — Question 1. Each of the following can be written in at least one of the forms $R\sin(\theta \pm \alpha)$ or $R\cos(\theta \pm \alpha)$, where $R > 0$ and $0° < \alpha < 90°$. Write down the exact value of R in each case:

a) $8\sin\theta + 15\cos\theta$
b) $12\sin\theta - 35\cos\theta$
c) $14\cos\theta - 20\sin\theta$
d) $2\sqrt{2}\cos\theta + 5\sqrt{2}\sin\theta$

2.18 — Question 2. State the maximum and minimum values of each of the following expressions:

a) $9\sin\theta - 40\cos\theta$

b) $10\sin\theta + 24\cos\theta$

c) $6\sqrt{3}\cos\theta + 4\sqrt{3}\sin\theta$

d) $3\sqrt{5}\cos\theta - 5\sqrt{3}\sin\theta$

2.18 — Question 3. State the maximum and minimum values of each of the following expressions:

a) $12 + 220\cos\theta - 21\sin\theta$

b) $50 - \cos\theta + 18\sin\theta$

c) $(16\cos\theta + 63\sin\theta)^2$

d) $(5 + 4\sin\theta - 9\cos\theta)^2$

2.18 — Question 4. State the maximum and minimum values of each of the following expressions:

a) $\dfrac{1}{14+5\sin\theta+12\cos\theta}$

b) $\dfrac{15}{90-84\sin\theta-13\cos\theta}$

c) $\dfrac{100}{(6+\sin\theta+\cos\theta)^2}$

d) $\dfrac{625}{90+(-33\cos\theta+56\sin\theta)^2}$

2.18 — Question 5. The maximum value of the expression $\sqrt{10+3\sqrt{3}\cos\theta+k\sin\theta}$ is 4. Find the possible values of k.

2.18 — Question 6. Write the following in the form $R\sin(x+\alpha)$ where $R>0$ and $0°<\alpha<90°$.

a) $3\sin x+3\cos x$
b) $5\sin x+5\sqrt{3}\cos x$
c) $9\sqrt{3}\sin x+9\cos x$
d) $4\sqrt{2}\sin x+2\sqrt{2}\cos x$

2.18 — Question 7. Write the following in the form $R\sin(x-\alpha)$ where $R>0$ and $0<\alpha<\frac{\pi}{2}$.

a) $2\sqrt{2}\sin x-2\sqrt{2}\cos x$
b) $3\sin x-4\cos x$
c) $12\sin x-5\cos x$
d) $11\sin x-\cos x$

2.18 — Question 8. Write the following in the form $R\cos(x+\alpha)$ where $R>0$ and $0°<\alpha<90°$.

a) $4\cos x - 4\sqrt{3}\sin x$
b) $5\cos x - \sin x$
c) $20\cos x - 30\sin x$
d) $13\cos x - 6\sin x$

2.18 — Question 9. Write the following in the form $R\cos(x-\alpha)$ where $R>0$ and $0<\alpha<\frac{\pi}{2}$.

a) $20\cos x + 99\sin x$
b) $56\cos x + 33\sin x$
c) $25\sqrt{3}\cos x + 75\sin x$
d) $2\sqrt{5}\cos x + 5\sqrt{5}\sin x$

2.18 — Question 10. Write each of the following in an appropriate harmonic form: $R\sin(\theta \pm \alpha)$ or $R\cos(\theta \pm \alpha)$, where $R > 0$ and $0° < \alpha < 90°$.

a) $6\cos\theta - 8\sin\theta$
b) $7\cos\theta + 24\sin\theta$
c) $5\sin\theta - 5\cos\theta$
d) $8\sqrt{2}\sin\theta + 10\sqrt{2}\cos\theta$

2.18 — Question 11. Find the maximum value of the following expressions, and the smallest positive value of θ, in degrees, for which this maximum occurs:

a) $\sin\theta + 5\cos\theta$
b) $9\cos\theta - \sqrt{2}\sin\theta$
c) $6\sin\theta - 4\cos\theta$
d) $10\cos\theta + 20\sin\theta$

2.18 — Question 12. Solve the following trigonometric equations:

a) $4\sqrt{2}\sin\theta - 4\sqrt{2}\cos\theta = 7,\ 0° < \theta < 360°$

b) $4\sin\theta + 2\cos\theta = 3,\ 0 < \theta < 2\pi$

c) $6\sqrt{3}\cos 3\theta + 5\sin 3\theta = 1,\ 0° < \theta < 240°$

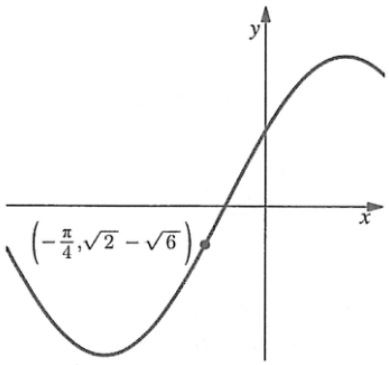
d) $10\sqrt{5}\cos 2\theta - 12\sqrt{5}\sin 2\theta = 5,\ 0 < \theta < 2\pi$

2.18 — Question 13. Describe a sequence of two graph transformations, in radians, that would map $y = \sin x$ onto the image $y = 4 + 10\sin x - \sqrt{2}\cos x$.

2.18 — Question 14. Write $40\cos 5x - 42\sin 5x$ in the form $R\cos(5x+\alpha)$ where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

2.18 — Question 15. Write $8\cos\frac{1}{2}x + 3\sin\frac{1}{2}x$ in the form $R\cos(\frac{1}{2}x - \alpha)$ where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

2.18 — Question 16. The diagram below shows a curve with equation $y = p\sin x + q\cos x$, where p and q are positive constants. The curve passes through the point $(-\frac{\pi}{4}, \sqrt{2} - \sqrt{6})$. It is given that the minimum value of the curve is -4. Find the exact values of p and q.



2.18 — Question 17. For a land-based wind turbine, the height above the ground, h metres, of the tip of one of the wind turbine blades after t seconds is modelled by the equation

$$h = k + 28\cos 4t + 54\sin 4t.$$

a) Before the wind turbine is switched on and starts moving, the blade tip in question is locked into a position 117 metres above the ground. Find the value of k.
b) The wind turbine is switched on. Find the maximum height the blade tip reaches above the ground as it rotates, and the time it takes the blade tip to first reach this position.

2.18 — Question 18. The depth of water, D meters, at a point in a tidal river is modelled by the equation

$$D = 4 + \sin(\tfrac{\pi}{6}t - 2) + 1.5\cos(\tfrac{\pi}{6}t - 2)$$

where t is the number of hours after midnight.

a) Find the minimum depth of the river at the point being measured.

b) A ferry can only cross the river when the depth of the water is at least 5 metres. Derek arrives at the ferry at 1pm. How long must he wait before the ferry starts operating again?

2.18 — Question 19. An engine piston moves up and down in a vertical cylinder. The point A is at the top of the piston. The vertical displacement, h mm, of A from its lowest position is modelled by the equation

$$h = 50 + 40\cos 20t - 30\sin 20t$$

where t is the time in seconds after the piston starts moving.

a) Find the maximum displacement of A.

b) In a single cycle of the engine, A is more than 80 millimetres above its lowest position for T seconds. Find the value of T.

c) The engine's speed is increased so that the piston now completes 15 cycles per second. Explain how the equation of the model should be modified to reflect this change in speed.

2.18 — Question 20.

a) Express $\sqrt{129}\cos\theta - 20\sin\theta$ in the form $a\cos(\theta + b)$, where $a > 0$ and $0 < b < 90°$.

A meteorologist studies the temperature and humidity in Saskatoon over one year. The temperature, $T°$C, is modelled by the equation

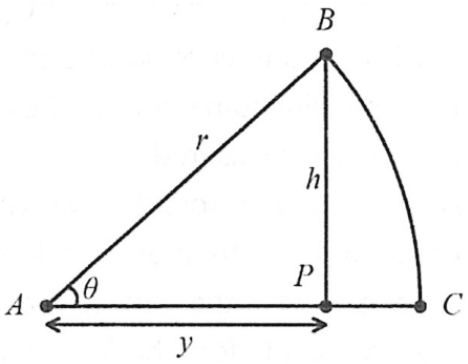
$$T = \sqrt{129}\cos(30t)° - 20\sin(30t)° + C$$

where t is the time in months after the start of the year.

a) Given that during the year, the maximum temperature in Saskatoon is 25°C, find the value of C.

b) The humidity, $H\%$, during the same year is modelled by $H = 10\cos(30t + 45)° + 72$. Find the temperature in Saskatoon when the humidity is at its maximum.

2.19 — Question 1. The diagram below shows a sector of a circle, ABC. By considering trigonometric ratios, show that as $\theta \to 0$ that $\sin\theta \approx \tan\theta \approx \theta$



2.19 — Question 2. Show that if $\sin\theta \approx \theta$ then as $\theta \to 0$, $\cos\theta \approx 1 - \frac{\theta^2}{2}$

2.19 — Question 3. When θ is small, find an approximation for the following:

a) $\theta \sin\theta$
b) $\cos\theta - 1$
c) $\cos 6\theta$
d) $\tan\theta - 2\sin\theta$
e) $\theta \tan 3\theta + \cos 3\theta$

2.19 — Question 4. When θ is small enough for θ^3 to be ignored, find an approximation for the following:

a) $\frac{\theta \tan \theta}{1-\cos \theta}$

b) $\cos \theta \cos 3\theta$

c) $\cos^2 \theta$

d) $\sin^2 \theta + \cos^2 \theta$

e) $\frac{\theta^2 \sin \theta}{1-\cos 2\theta}$

f) $\frac{\cos 8\theta - \cos 4\theta}{\sin 8\theta - \sin 4\theta}$

g) $\frac{1-\cos 2\theta}{6 \tan \theta}$

2.19 — Question 5.

a) Find an approximation for $\cos 0.4$ by using $\theta = 0.1$ and small angle approximations.

b) Find the percentage error of your approximation.

2.19 — Question 6. When θ is small, use the standard small angle approximations for $\sin\theta$ and $\tan\theta$ find:

a) $\theta \operatorname{cosec}\theta$

b) $\cot 2\theta$

c) $\cot 4\theta - \operatorname{cosec} 2\theta$

2.19 — Question 7.

$$\mathrm{f}(x) = \sec x(\sin x + \cos x)$$

a) By using the small angle approximations for $\tan x$ show that $\mathrm{f}(x) \approx 1 + x$

b) By using the small angle approximation for $\sin x$ and an alternative approximation, $\cos x \approx 1$, show that $\mathrm{f}(x) \approx 1 + x$

c) Find the percentage error when approximating $\mathrm{f}(0.1)$

2.19 — Question 8.

a) When θ is small, show that $-2 + \cos\theta + \sin^2\theta \approx -1 + \frac{1}{2}\theta^2$

b) Hence find an approximate solution to $-2 + \cos\theta + \sin^2\theta = 4\tan\theta$

2.19 — Question 9. Given x is small, find the approximate solution to

$$\frac{1 - \cos(\frac{1}{10}x)}{\sin x} = \frac{1}{5} - \tan x$$

2.19 — Question 10.

a) Find, for small values of x, an approximation for $\mathrm{f}(x) = 2 + 4\tan\frac{x}{2} - 8\sin\frac{x}{3}$

b) Use your result in part a) to sketch an approximation for the graph of $\mathrm{f}(x)$ for $-0.2 \leq x \leq 0.2$. Include the start and end coordinates in your sketch.

2.19 — Question 11. Given that the root of the equation is small, find the single solution to $2x - \frac{1}{4} + \cos x = 0$

2.19 — Question 12. Given x is small enough for x^3 to be ignored, by approximating $\tan x + \tan x \cos 4x$ show that

$$\int_0^{0.3} \sqrt[3]{\tan x(1+\cos 4x)}\, dx \approx 0.190$$

2.19 — Question 13.

a) When θ is small, find an approximation to $\cos\theta - \cos 2\theta$

b) Hence find $\lim_{\theta\to 0} \frac{\cos\theta - \cos 2\theta}{\theta^2}$

2.19 — Question 14. Find these limits:

a) $\lim_{h\to 0} \frac{\cos h - 1}{h}$

b) $\lim_{h\to 0} \frac{\sin h}{h}$

2.19 — Question 15. Prerequisite knowledge for Q15-Q18: Differentiation from First Principles and Compound Angle Formulae Differentiate from first principles the following functions by using knowledge about small angle approximations:

a) $\mathrm{f}(\theta) = \sin\theta$

b) $\mathrm{f}(\theta) = \cos\theta$

2.19 — Question 16. Using differentiation from first principles, for $f(x) = \cos(x)$, find the exact value of $f'(\frac{\pi}{3})$

2.19 — Question 17. Prove, using differentiation from first principles, that $\frac{d}{dx}(\tan\theta) = \sec^2\theta$

2.19 — Question 18. Prove, using differentiation from first principles, that $\frac{d}{dx}(\sec\theta) = \sec\theta\tan\theta$

2.19 — Question 19. Prerequisite knowledge for Q19-Q20: Binomial Series

a) Find the first three terms, in ascending powers of x of the binomial expansion of $(1-8x)^{1/2}$

b) Hence, for small values of x, show that $\sin 2x + \sqrt{\cos 4x} \approx a + bx + cx^2$ where a, b and c are constants to be found and x is small enough for x^4 to be ignored.

2.19 — Question 20.

a) Find the first two terms, in ascending powers of x of the binomial expansion of $(1 - \frac{1}{2}x^2)^{1/3}$

b) Hence find an approximation for $\int_0^{0.2} \sqrt[3]{\cos x}\, dx$ giving your answer to 4 decimal places.

2.19 — Question 21. Prerequisite knowledge for Q21-Q22: Differentiation of Trigonometric Functions

Find the value of $\frac{dy}{dx}$ at the origin of $x = \frac{1}{2}\tan 4y$ by:

a) using Calculus

b) using a small angle approximation and Calculus

2.19 — Question 22. Find the value of $\frac{dy}{dx}$ when $x = 0$ for $x = 3\cos 16y$ where $0 < y < \frac{\pi}{2}$ by:

a) using Calculus

b) using a small angle approximation and Calculus

2.19 — Question 23. Prerequisite knowledge for Q23-Q24: Newton-Raphson Method Given x is small, find an approximate solution giving your answers to 3 significant figures for $\frac{1}{2} - x - \sin 2x = 0$

a) by using small angle approximation

b) by using a Newton-Raphson approximation starting at $x_0 = 0$

2.19 — Question 24. Given x is small, find an approximate solution giving your answers to 3 significant figures for $2 + 5x + \sin x = \cos x$

a) by using small angle approximation

b) by using a Newton-Raphson approximation starting at $x_0 = 0$